

Explainable Affirmative Action

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We study *Prioritized Selection Problems*, in which an organization selects individuals based on observable characteristics and a priority ranking. We introduce *outcome-based selection rules*, which are defined by a collection of feasible selections and a greedy processing algorithm.

We show that outcome-based rules are the only rules that *respect improvements*, are *consistent*, and are *lower invariant* (Theorem 1). Additionally, given a collection of feasible selections, these rules choose the unique selection that *respects priorities* (Proposition 1).

Some existing algorithms (such as “minimum guarantee” reserves and “smart reserves”) implement outcome-based selection rules, while others (such as “over and above” reserves) do not. We introduce a new algorithm for reserve matching based on “maximal reserves”, which optimizes over the outcomes chosen by slot-specific priority algorithms from prior work (Proposition 2). We also identify conditions under which slot-specific priority algorithms are outcome-based (Proposition 3), and show that *any* outcome-based rule can be implemented using maximum quotas (Proposition 4).

Finally, we consider the Chilean Constitutional Assembly election, in which the elected representatives had to include an appropriate number of candidates from each party and gender. We show that the implemented rule corresponds to the outcome-based selection rule associated with these constraints (Theorem 2).

1 INTRODUCTION

We study *Prioritized Selection Problems*, in which an organization is presented with a set of individuals, and must choose which to accept. We assume that the organization has a complete priority ranking of individuals, which may be determined by various factors, including date of application, test scores or other measures of merit, a lottery, or some combination of these. The organization also has objectives or constraints that are not reflected in the priority ranking, but influence the final selection. Examples of such settings include:

- **Chilean Constitutional Assembly Election.** In 2021, Chile held an election to choose 155 representatives to rewrite its constitution. Candidates were prioritized by votes received. However, the elected representatives from each district had to include an equal number of men and women, and include an appropriate number of members of each party (based on that party’s vote share in the district).
- **US Immigrant Visa Programs.** H1B and Diversity visas are allocated by lottery. However, some H1B visas are reserved for applicants with graduate degrees from US institutions, and the Diversity visa lottery limits the number of successful individuals from each country.
- **Affordable Housing Allocation.** Applicants for affordable housing through New York City’s “HousingConnect” website are placed in a uniformly random order. However, a certain number of units must go to people with disabilities, city employees and residents of the local community district. Furthermore, units have different eligibility criteria based on household size and income.
- **School Assignment.** Specialized high schools in New York City admit students based on a test score, with some seats reserved for students from disadvantaged backgrounds [Faenza et al., 2022]. In many countries, priority for university admissions is determined by exam scores, with additional considerations. In India, positions are reserved for people from disadvantaged castes, people with disabilities, women, and people from the local state [Baswana et al., 2019]. In Brazil, positions are reserved for students who attended public high school, are low-income, or belong to a disadvantaged minority group [Aygun and Bo, 2019].

- **Government Positions.** In India, priority for government positions is determined by exam scores, but positions are reserved for applicants from marginalized groups, including women and “backwards classes.” [Aygün and Turhan, 2022, Sönmez and Yenmez, 2022].

It is not obvious how to best *combine* the priority ranking and these additional goals to reach a final selection. In practice, there are many ad hoc methods for doing so, but these algorithms frequently produce outcomes that can be difficult to explain to participants. For example,

- **Affordable Housing:** In New York, applicants from outside the community district are often passed over for lower-ranking applicants from the community district, even if the community preference constraint is not binding [Arnosti, 2022]
- **Specialized High Schools:** high scoring students who are eligible for the Discovery Program are sometimes assigned to their second or third choice, while seats in their preferred schools are offered to lower-scoring students [Faenza et al., 2022].
- **University Admissions:** In Brazil, cutoff scores for applicants claiming multiple privileges are frequently *higher* than corresponding cutoffs for applicants claiming a subset of those privileges [Aygün and Bo, 2019].
- **Civil Service:** Numerous lawsuits in India document cases where the algorithm selected a woman from a majority group instead of another woman from a backwards class *with a higher exam score* [Sönmez and Yenmez, 2022].

Each of these examples features an apparent violation of priority that is difficult to justify on the basis of the diversity goals. Individually, each example provides an interesting research opportunity, and the papers mentioned above generally provide “solutions” (new algorithms) for the particular context they study. Collectively, however, these examples illustrate the challenge of implementing affirmative action and diversity policies effectively.

Our primary contributions are to introduce the family of outcome-based selection rules and to show that these rules are characterized by three axioms (respecting improvements, lower invariance, and consistency). These rules eliminate ad-hoc priority violations, ensuring that any deviation from the priority ranking is necessitated by the diversity constraints. Second, we compare our approach to various policies that use reserves and quotas, and identify existing rules that do – and do not – implement outcome-based rules. Finally, we show that the rule used in Chile’s Constitutional Assembly is outcome-based. We now elaborate on each contribution.

1.1 Axiomatizing Explainability

Given a set of individuals $\mathcal{I}$, a *selection* is a subset of these individuals, and a *selection rule* φ is a function mapping each priority order $>$ to the resulting selection $\varphi(>)$. We consider three natural properties which constrain how the rule responds to changes in the priority list.

- **Respects improvements:** improving an individual’s priority never harms that individual.
- **Lower invariant:** whether an individual is selected depends only on the priorities of higher-priority individuals, and not on those of lower-priority individuals.
- **Consistent:** if a change in an individual’s priority does not change that individual’s outcome, then it does not change the final selection.

Theorem 1 establishes that a selection rule satisfies all three of these properties if and only if it is *outcome-based*. Outcome-based selection rules are defined by a collection of feasible selections $\mathcal{F} \subseteq \{0, 1\}^{\mathcal{I}}$ along with a greedy processing rule presented in Algorithm 1. This rule considers individuals in priority order, and commits to selecting each individual if it is feasible to do so while honoring the commitments to higher-priority individuals.

We use the term “outcome-based” because the public and policymakers typically express preferences over acceptable *outcomes* rather than the *algorithms* used to derive them. In many of the

	Selection Rule 1	Selection Rule 2
<i>Algorithmic Description</i>	Hire top woman, then Hire top remaining candidate.	Hire top candidate, then Hire top remaining woman.
<i>Outcome Description</i>	Selections must have 2 people, at least 1 woman, and must respect priorities.	

Fig. 1. Outcome-based selection rules provide a new way to describe existing procedures. For example, Theorem 2 establishes that the algorithm used to select members of Chile’s Constitutional Assembly is equivalent to an outcome-based rule, despite having a very different description. A simpler example is Selection Rule 1 (a “minimum guarantee” reserve), which is equivalent to the outcome-based rule associated with the constraints that selections must contain at least one woman and at most two people. Other rules, such as Selection Rule 2 (an “over and above” reserve), do not have any outcome-based description.

applications mentioned earlier, both priorities and feasible outcomes are clearly specified, but the algorithm for combining them is either unspecified or receives very little attention. For example, Gonczarowski et al. [2019] describe a matching system in which

Each PMA [Pre-Military Academy] defines a set of “populations” that it cares about (e.g., based on gender, religiousness, city of origin, belonging to certain minority groups). For each such population the PMA is allowed to define a maximum quota as well as minimum target. In addition to ranking the candidates... the PMA also specifies [a] set of populations to which each candidate belongs.

They note that this information does not identify a unique selection rule (“There are several issues that need be specified before this becomes fully formal”), and comment that “virtually none of the PMAs showed any desire to dig into these issues.” Relatedly, Pathak et al. [2022] run experiments showing that people understand the consequences of changing the number of positions reserved for different categories, but do not understand the consequences of changing the algorithm (processing order) used to fill these positions.

To illustrate that the feasible selections $\mathcal{F}$ and the priority order $>$ do not obviously pin down the final selection, we present the following example.

EXAMPLE 1. *There are four individuals $1 > 2 > 3 > 4$. The feasible selections are $\mathcal{F} = \{\{1, 4\}, \{2, 3\}\}$.*

These feasible selections arise naturally if we must select 2 applicants while including 1 woman, 1 man, 1 liberal and 1 conservative. (This example is inspired by the Chilean Constitutional Assembly discussed later in this paper.) If candidates 1 and 2 are women (3 and 4 are men) and candidates 1 and 3 are liberal (2 and 4 are conservative), then the feasible selections are exactly $\{1, 4\}$ and $\{2, 3\}$.

In this case, it is not clear whether $\{1, 4\}$ or $\{2, 3\}$ should be chosen. However, we believe that $\{1, 4\}$ is easier to explain. We formalize this argument as follows. We say that a feasible selection *respects priorities* if, for every individual who is not selected, it is impossible to select this individual without displacing some higher-priority individual who is currently selected. Proposition 1 establishes that there is always a unique feasible selection that respects priorities ($\{1, 4\}$ in Example 1), and it can be found by the greedy procedure described in Algorithm 1.

We can explain to individuals 2 and 3 that selecting them would require displacing the higher-priority individual 1. No such simple explanation is available if we choose $\{2, 3\}$. In that case, we must explain to individual 1 why they were not chosen despite having the top priority. Our explanation will be complicated by the fact that had the bottom of the ranking been slightly different

(with individual 4 ahead of individual 3), we presumably would have selected $\{1, 4\}$. Individual 1 may feel that the precise order of lower-priority agents should be irrelevant, motivating our definition of lower invariance.

1.2 Special Cases: Reserves and Quotas

Having introduced outcome-based selection rules and characterized them as uniquely explainable, we show how these rules can capture diversity and affirmative action considerations that arise in practice. We consider two ways to define feasible selections: *reserves* and *quotas*.

When using reserves, the feasible selections are defined by a set of positions and a compatibility graph G indicating which individuals may be matched to which positions. We consider several ways to interpret the graph G . *Hard reserves* require that compatibility constraints are always strictly enforced. *Soft reserves* allow the selection of incompatible individuals so long as all compatible individuals have already been selected. We formulate two interpretations of this requirement: *soft maximal reserves* require that feasible selections contain all individuals matched in some maximal matching M of G . *Soft maximum reserves* are more restrictive: they require that M be a maximum matching. The “meritorious horizontal choice rule” of Sönmez and Yenmez [2022] (called “smart reserves” by Pathak et al. [2024]) coincides with the outcome-based rule that uses maximum reserves. Meanwhile, the selections that can be chosen by the outcome-based rule that uses maximal reserves are exactly those that can be found by the slot-specific priority (SSP) algorithms considered by Pathak et al. [2024] (see Proposition 2). Importantly, however, SSP algorithms are *not* outcome-based, and often choose a selection that does not respect priorities. Proposition 3 establishes conditions under which SSP algorithms implement outcome-based rules. One consequence of this result is that “minimum guarantee” reserves are outcome-based.

Quotas, meanwhile, are specified by (i) a finite set of traits T ; (ii) for each individual a subset of traits that the individual possesses; and (iii) upper and lower bounds (quotas) on the number of selected individuals with each trait. Selected individuals contribute towards quotas for all traits that they possess. Proposition 4 establishes a universality result for quotas, proving that a system of upper quotas is expressive enough to replicate *any* outcome-based selection rule, though doing so may require a number of traits that grows exponentially in the number of individuals. In general, determining the existence of a feasible selection is NP-complete (Proposition 7). However, if the traits form a hierarchy, then the associated outcome-based rule can be computed in polynomial time, and produces a feasible outcome that priority dominates all others (Proposition 5). Furthermore, the hierarchy condition is “necessary” in a maximal domain sense. In cases where traits do not form a hierarchy (such as that considered by Gónczarowski et al. [2019]) our outcome-based rule guarantees selection of a feasible outcome that respects priorities, and hence is not priority dominated.

1.3 Electing the Chilean Constitutional Assembly

In 2021, the country of Chile convened an assembly to propose a new constitution. Members of this assembly were selected by a special election. Each district elected a set of representatives whose parties needed to reflect the district’s vote totals. Furthermore, representatives needed to be equally balanced between male and female. The algorithm used by Chile begins by ignoring the gender constraints, and then corrects imbalances by replacing the elected candidates from the over-represented gender who receive the fewest votes. We show that this algorithm is equivalent to the outcome-based rule associated with the party and gender constraints (Theorem 2). This implies that for each candidate who was not elected, the only way to select them while complying with gender and party constraints is to reject a candidate who received more votes. This fact is

not obvious from the description of the algorithm, but could help to explain the final outcome to candidates and voters.

2 RELATED WORK

Our paper contributes to a large and growing literature on assignment and matching problems with distributional constraints. There are too many papers to discuss each one in detail, so we divide our discussion into two clusters: those where individuals belong to disjoint groups, and those where affirmative action targets overlapping groups. Within each cluster, we focus our discussion on the most related work.

To the extent possible, we have attempted to align our language with that used by prior work. However, what we call “minimum quotas” have been referred to by many different names. Hafalir et al. [2013], Echenique and Yenmez [2015], and Abdulkadiroglu and Grigoryan [2021] use the term “quotas” exclusively for maximums, and the term “reserves” for minimums. Fragiadakis and Troyan [2017] refer to minimums as a “floor” or a “lower quota”. Similarly, Tomoeda [2018] uses “floor” and “minimum quota” interchangeably. These papers consider settings where each individual has exactly one trait. In such cases, our definitions of minimum quotas and reserves coincide, so our terminology can be seen as consistent with all of these papers.

Among papers that allow individuals to have multiple traits, there is also no consensus on how to refer to what we call minimum quotas. Sönmez and Yenmez [2022] and Aygün and Turhan [2026] refer to them as “one to all reserves.” Gonczarowski et al. [2019] use the term “minimum targets,” to reflect the fact that their algorithms may fail to meet these targets (even if it is possible to do so). Our terminology is most aligned with the literature on selecting representative panels for citizens’ assemblies, which uses the term “lower quotas” [Assos et al., 2025, de Azevedo and Gözl, 2026, Flanigan et al., 2021a,b, 2024].

A number of papers consider settings with many institutions (i.e. universities or hospitals), and constraints on the total number of seats that can be filled at various combinations of institutions, rather than on the types of selected applicants (see Biró et al. [2010], Kamada and Kojima [2018], and references therein). They call these constraints upper or lower quotas. This terminology is consistent with ours after a transformation that maps their *seats* to our *individuals*. Then their quotas on the number of filled seats from various institutions correspond to our quotas on the number of selected individuals with various traits.

In the following discussion, we use our language to describe the results from prior work.

2.1 Disjoint Types

Most early matching work on distributional constraints and affirmative action assumes that individuals can be partitioned into types, and that constraints are specified as quotas on the number of selected individuals of each type.

Abdulkadiroğlu and Sönmez [2003] and Abdulkadiroğlu [2005] assume only maximum quotas. Kojima [2012] shows that a maximum quota on “majority” students can end up harming all “minority” applicants. Hafalir et al. [2013] also consider a model with majority and minority students, and argue that minority reserves (minimum quotas) result in more efficient outcomes than majority (maximum) quotas. A related model of Ehlers et al. [2014] attempts to achieve diversity goals by imposing both minimum and maximum quotas on the number of students of each type. Not surprisingly, if these are treated as hard constraints, a feasible assignment of students to schools may not exist; so these authors interpret lower bounds as “soft” constraints that should be satisfied to the extent possible.

Doğan and Yildiz [2023] and Abdulkadiroglu and Grigoryan [2021] also provide characterization results for prioritized selection problems with constraints. Like our work, these papers assume existence of a baseline priority ranking, in addition to other diversity objectives. Both papers

assume these objectives are specified as quotas over non-overlapping subsets of applicants, whereas we allow much more generality.

If we model the setting of Doğan and Yildiz [2023] using our framework, their “monotonicity in priority improvements” axiom is weaker than the combination of our “respecting improvements” and “consistency” axioms. As a result, only some of the selection rules characterized by their axioms are outcome-based. In particular, an “over and above” implementation of minority reserves satisfies all of their axioms, but is not consistent.

Because Abdulkadiroglu and Grigoryan [2021] assume that agents can be partitioned into disjoint types, there is a feasible selection that priority dominates all others (see our Proposition 5). They provide an explicit algorithm (the “regular reserves rule”) to find this selection, which is equivalent to our outcome-based rule for their specific setting.

Recent work by Pathak et al. [2022] loosely relates to our motivation of explainability. It uses a survey of over 1,000 people to demonstrate that the effects of changing the processing order of different positions are poorly understood. This helps to reinforce our belief that people are generally better at understanding and forming opinions about final outcomes, rather than the procedures that generate them.

2.2 Overlapping Types

An important shortcoming of these early models is the assumption that agents are partitioned into disjoint types. In practice, affirmative action policies often target groups whose memberships overlap. A number of recent papers consider this case (as do we). Most of these papers focus on a specific application, and describe feasibility constraints, axioms, and algorithms suitable for that domain. For this reason, it can be difficult to translate results from one paper to the next. One of the goals of our work is to propose axioms and a greedy algorithm which are general enough to be applied regardless of the structure of the feasible sets.

Perhaps the paper whose approach is most similar to ours is Hafalir et al. [2026]. They consider choice rules that determine a final selection based on (i) a set of applicants, and (ii) a priority ranking for all individuals. Fixing the priority ranking, these rules determine a choice function. Fixing the set of applicants, these rules determine a selection rule (as defined in this paper). The family of rules on which they focus is “Diversity Choice Rules,” which rely on a function f that assigns a “diversity score” to each selection. For any set of applicants $\mathcal{I}$, their feasible sets are the most diverse subsets of $\mathcal{I}$ (that is, $\mathcal{F} = \arg \max_{S \subseteq \mathcal{I}} f(S)$). They select among $\mathcal{F}$ using a greedy processing algorithm. Thus, our outcome-based selection rules can be seen as a special case of their Diversity Choice Rule where the set of applicants $\mathcal{I}$ is fixed and known.

Although their paper considers a generalization of our outcome-based selection rules, our central questions are fairly different. They focus primarily on identifying conditions on f to ensure that $\mathcal{F}$ induces a matroid. The conditions they provide are based on discrete concavity notions, making them possibly more general but also much more abstract than those in our Proposition 5. By contrast, our focus is on explainability. We provide two characterization results showing that outcome-based rules are uniquely explainable, regardless of whether the underlying constraint set induces a matroid.

Several papers present results that are closely related to our findings in Section 4. Gonczarowski et al. [2019] study a setting with lower and upper quotas, and identify the importance of having the traits form a hierarchy (which they refer to as ‘laminar’). However, in order for their algorithms to be guaranteed to find a priority dominant selection, their Theorems 1 and 2 require further restrictions on the set of traits with a nonzero lower quota. A result from Yokoi [2017], reproduced as our Proposition 5, establishes that these restrictions are unnecessary. When traits do not form a hierarchy, Gonczarowski et al. [2019] propose a computationally simple algorithm that does not

respect priorities, while we propose finding the selection that respects priorities (which is NP-hard in the worst case).

Aygün and Turhan [2022] and Sönmez and Yenmez [2022] study affirmative action policies in India. Whereas they focus on axioms motivated by the Indian Constitution, we attempt to formalize the notion of explainability. This difference leads us in different directions: their “vertical reservations” are constitutionally mandated, but are not consistent (and thus not outcome-based). However, as mentioned in the introduction, their analysis of overlapping “horizontal reservations” leads them to propose the “meritorious horizontal choice rule”, which coincides with the outcome-based rule associated with maximum reserves.

Cembrano et al. [2021] compare the selection rule used for the Chilean Constitutional Convention to many alternatives. However, they do not provide any characterization of this rule. By contrast, we show that this procedure finds the unique selection that respects priorities.

Imamura and Kawase [2022] consider a setting with multiple institutions, in which each institution considers only certain subsets of individuals to be feasible. They note, as we do, that greedy selection with feasibility constraints may sometimes require solving NP-complete problems.

3 EXPLAINABLE SELECTION RULES

In a *selection problem* there is a set of applicants $\mathcal{I}$, from which a decision maker has to select a subset $S \subseteq \mathcal{I}$. There is a strict priority order $>$ over $\mathcal{I}$. We denote by $\mathbf{P}$ the set of all priority orders over $\mathcal{I}$. A *selection rule* is a mapping $\varphi : \mathbf{P} \rightarrow 2^{\mathcal{I}}$ that determines a selection for every priority order. We let $\varphi(>)$ denote the selection of rule φ for priority order $>$.

3.1 Respecting Priorities

Our first characterization result is for cases where there is an explicitly defined collection of feasible selections $\mathcal{F} \subseteq 2^{\mathcal{I}}$. This may capture capacity constraints, diversity goals or other restrictions imposed by law or by the designer. In this section, our goal is to define a selection rule that (i) always returns a feasible selection and (ii) follows the priority order “as much as possible.” To formalize this second objective, we introduce a method to compare feasible selections.

DEFINITION 1. *Selection S' **priority dominates** selection S if (and only if) $S' \neq S$ and there exists an injective function ψ from S to S' , such that for each $i \in S$, $\psi(i) \geq i$. Equivalently, S' **priority dominates** S if (and only if) S' selects at least as many “top k priority” individuals for every $k \in \mathbb{N}$.*

If $1 > 2 > 3 > 4$, then $\{1, 3\}$ priority dominates $\{1, 4\}$, $\{2, 3\}$, and $\{2, 4\}$. However, neither of the selections $\{1, 4\}$ and $\{2, 3\}$ priority dominate the other.

Priority domination induces a natural partial order on sets of applicants, which has appeared in other papers under different names. For example, Aygün and Turhan [2023] refer to this as “merit-based domination”, while Sönmez and Yenmez [2022] use the name “Gale domination.” Gale [1968] simply calls a selection that priority dominates all others “optimal.” Abdulkadiroglu and Grigoryan [2021] offer an alternative (stronger) definition of “priority domination”.¹

If our goal is to respect the priority order as much as possible, then at a minimum, we should choose a feasible selection that is not priority dominated. However, this does not identify a unique outcome if there are multiple undominated feasible selections, as in the case of Example 1. To resolve this question, we propose the following criteria.

¹Their definition (which we will call AG-domination) is that S' AG-dominates S if every agent in $S' \setminus S$ has higher priority than every agent in $S \setminus S'$. This definition has the counter-intuitive property of not being transitive: if $1 > 2 > 3 > 4$, then $\{1, 3\}$ AG-dominates $\{1, 4\}$ and $\{2, 3\}$, both of which AG-dominate $\{2, 4\}$, but $\{1, 3\}$ does not AG-dominate $\{2, 4\}$.

Algorithm 1: Outcome-Based Selection

Input: Set of applicants $\mathcal{I}$, feasible selections $\mathcal{F}$ and priority order $>$, with $i_1 > i_2 > \dots > i_n$.

$S := \emptyset$;

foreach $k = 1, \dots, n$ **do**

if $\exists S' \in \mathcal{F} : S \cup \{i_k\} \subseteq S'$ **then**

$S = S \cup \{i_k\}$

end

end

return S

DEFINITION 2. A feasible selection $S \in \mathcal{F}$ **respects priorities** if, for each unselected applicant $i \notin S$, it is impossible to select them without displacing a higher-priority applicant. Formally, there is no feasible selection $S' \in \mathcal{F}$ such that $\{i\} \cup \{j \in S : j > i\} \subseteq S'$.

Respecting priorities is intended to capture the absence of “justified envy” [Abdulkadiroğlu and Sönmez, 2003]. In simple settings with only a global capacity constraint, individual i has justified envy for j if i has higher priority than j , yet j is selected while i is not. In settings with richer constraints, there are several plausible definitions for what it means for i ’s envy for j to be justified. One might say that i ’s envy is only justified if replacing j with i maintains feasibility. However, if the feasible sets are $\{1, 2\}$ and $\{3, 4\}$ and the priority order is $1 > 2 > 3 > 4$, then this interpretation would imply that selecting $\{3, 4\}$ does not produce any justified envy. We instead take the position that i ’s envy for j should be considered justified whenever it is possible to take i without harming any individuals with higher priority than i (but perhaps harming j and many other lower-priority individuals).

Delacrétaz et al. [2023] model refugee assignment as a matching problem in which each locality has feasibility constraints defined by a multi-dimensional knapsack problem. If we define our feasible sets $\mathcal{F}$ as they do, then our definition of respecting priorities coincides with theirs, which they call “interference-freeness.”

In addition, we point out that our definition of respecting priorities offers a “simple” explanation for each individual’s outcome. Suppose that an individual i complains that they were not selected, while some lower priority individuals were. If we have chosen a selection that respects priorities, then we can tell i that these lower priority individuals were not the reason that i was not selected. Rather, i was not selected because any feasible selection including i would require the displacement of some individual with a *higher* priority.

Applying this concept to Example 1 with $1 > 2 > 3 > 4$ and $\mathcal{F} = \{\{1, 4\}, \{2, 3\}\}$, the selection $\{1, 4\}$ respects priorities, while the selection $\{2, 3\}$ does not. We interpret this as saying that if we choose $\{2, 3\}$, there is no “simple” explanation for why individual 1 was not selected. The following (straightforward) result establishes that there is always a unique selection that respects priorities.

PROPOSITION 1. For any non-empty feasible set $\mathcal{F}$ and any priority order $>$, let $OB_{\mathcal{F}}(>)$ denote the outcome of Algorithm 1. Then:

- (i) $OB_{\mathcal{F}}(>)$ is not priority dominated by any other feasible selection.
- (ii) $OB_{\mathcal{F}}(>)$ is the unique feasible selection that respects priorities.

PROOF. Fix $\mathcal{F}$ and $>$. Let S be any feasible selection other than $OB_{\mathcal{F}}(>)$. Let i be the highest priority individual who receives a different outcome under $OB_{\mathcal{F}}(>)$ and S . By assumption, S and $OB_{\mathcal{F}}(>)$ make identical decisions on individuals with higher priority than i : $\{j : j > i\} \cap S = \{j : j > i\} \cap OB_{\mathcal{F}}(>)$. Note that if $i \in S$, then $i \in OB_{\mathcal{F}}(>)$ by definition: when $OB_{\mathcal{F}}(>)$ considered

individual i , S was a feasible selection containing i and all higher-priority individuals that had already been selected, so $OB_{\mathcal{F}}(>)$ must have selected i . Because S and $OB_{\mathcal{F}}(>)$ result in different outcomes for i , it must be that $i \in OB_{\mathcal{F}}(>)$ but $i \notin S$. This immediately implies that (i) S does not priority dominate $OB_{\mathcal{F}}(>)$, and (ii) S does not respect priorities, as $OB_{\mathcal{F}}(>)$ is feasible and selects i along with all higher-priority individuals selected by S . $\square$

3.2 A Characterization: Respecting Improvements, Lower Invariance, and Consistency

In practice, selection rules need not be defined through a set of feasible outcomes. Often, they are defined by a description of an algorithm for considering and selecting applicants. In such cases, the feasible outcomes are often defined only implicitly. As one example, we consider the cases of “over and above” and “minimum guarantee” reserves [Pathak et al., 2022].

EXAMPLE 2. *There are four individuals, 1, 2, 3, 4. Individuals 1 and 3 belong to a minority group. We can select at most two individuals, and wish to reserve at least one seat for minorities. The “over and above” selection rule φ^{OA} first selects the highest-priority individual, and then selects the highest priority minority that remains. The “minimum guarantee” selection rule φ^{MG} first selects the highest-priority minority, and then selects the highest priority individual that remains.*

These rules are defined without explicitly identifying feasible selections. Natural questions are,

- (1) Can an outcome-based rule (with suitably chosen $\mathcal{F}$) reproduce the behavior of φ^{OA} and φ^{MG} ?
- (2) If not, why not? What properties do outcome-based rules have, that other rules do not?

It turns out that φ^{MG} is equivalent to the outcome-based rule associated with the feasible selections

$$\mathcal{F} = \{\{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\}, \{3, 4\}\}.$$

(Note that $\{2, 4\}$ is never selected, as neither 2 nor 4 is a minority.) Meanwhile, we claim that **no** choice of $\mathcal{F}$ allows $OB_{\mathcal{F}}$ to reproduce the behavior of φ^{OA} . To see this, note that any such $\mathcal{F}$ must contain each of the five pairs listed above, as each of those pairs can be produced by some priority ordering. But when the priority order is $1 > 2 > 3 > 4$, φ^{OA} selects $\{1, 3\}$, even though this selection is priority dominated by $\{1, 2\}$.

Thus, while choosing $\mathcal{F}$ gives the designer substantial flexibility (as we explore in Section 4), restricting to the family of outcome-based rules does rule out certain selection rules commonly used in practice. One might therefore ask, why should a designer use an outcome-based rule? What advantages do these rules offer?

We show that outcome-based rules satisfy three desirable properties, and are in fact the *only* selection rules to satisfy all three properties. To define each of these properties, we introduce new notation. Given a priority order $>$ and a subset of individuals $S \subset \mathcal{I}$, we let $>_{-S}$ be the restriction of $>$ to $\mathcal{I} \setminus S$. In other words, for $i, j \in \mathcal{I} \setminus S$,

$$i >_{-S} j \iff i > j.$$

When $S = \{i\}$ is a singleton, we write $>_{-i}$ in place of $>_{-\{i\}}$.

Our first property is *respecting improvements*, which states that improvements to an individual’s priority should weakly improve their outcome.

DEFINITION 3. *A selection rule φ **respects improvements** if for any applicant $i \in \mathcal{I}$, and for every $>, >' \in \mathbf{P}$ such that (i) $>_{-i} = >'_{-i}$ and (ii) the priority of applicant i is higher in $>'$ than in $>$, then*

$$i \in \varphi(>) \Rightarrow i \in \varphi(>').$$

This is a very natural property, which has appeared in many prior papers, including Balinski and Sönmez [1999], Sönmez and Switzer [2013], Kominers and Sönmez [2016], Kominers [2020], and Biró et al. [2023].

The second property is *lower-invariance*, which says that whether an individual is selected should depend only on the priority of higher-priority individuals, and not on the exact priority of lower-priority individuals.

DEFINITION 4. A selection rule φ is **lower invariant** if for any applicant $i \in I$, and every $>, >' \in \mathbf{P}$ such that $>_{-\{j:i>j\}} = >'_{-\{j:i>'j\}}$,

$$i \in \varphi(>) \Leftrightarrow i \in \varphi(>').$$

Properties similar to lower invariance have been studied in the matching literature. For instance, Ehlers and Klaus [2014] study an axiom called *truncation invariance*, which says that if any student truncates their preference report at the school they are matched to, it should not affect the final outcome. Hashimoto et al. [2014] proposed a similar axiom for randomized mechanisms termed *weak invariance*: they require an agent’s share of an object to be independent of their ranking of less-preferred objects. In our setting, there are no preferences (as there is a single institution), and our lower invariance property concerns changes to the *priority order*. Lower-invariance is desirable because of its natural connection to explainability: if individual i is not selected, it is possible to provide i an explanation without referencing the priorities of lower-ranking applicants.

The final property is *consistency*, which says that a change to an individual’s priority which does not affect whether that individual is selected should also not affect who else is selected.

DEFINITION 5. A selection rule φ is **consistent** if for any applicant $i \in I$, and for every $>, >' \in \mathbf{P}$ such that $>_{-i} = >'_{-i}$,

$$\{i\} \cap \varphi(>) = \{i\} \cap \varphi(>') \Rightarrow \varphi(>) = \varphi(>').$$

We believe that rules which are not consistent can exhibit difficult-to-explain behavior. As an example, consider a selection rule φ which behaves as follows

$$\varphi(1 > 2 > 3 > 4) = \{1, 2, 3\},$$

$$\varphi(2 > 1 > 3 > 4) = \{1, 2, 4\}.$$

It seems difficult to explain why 3 was not selected in the second scenario. These scenarios seem almost indistinguishable from 3’s perspective (in both cases, 3 is ahead of 4, behind 1 and 2, and both 1 and 2 are selected), and yet any valid explanation must differentiate between them. Put differently, although φ sometimes selects the set $\{1, 2, 3\}$, it fails to do this even in some cases where these individuals occupy the first three positions on the priority list!

The behavior of φ may seem pathological, but it can easily arise from “slot-specific priority” algorithms considered in prior work, as the following example shows.

EXAMPLE 3. We have three positions A, B, C , each reserved for a targeted population. Position A is reserved for 1 or 2, position B for 1 or 3, and position C for 2 or 4. The algorithm φ first takes the highest-priority individual eligible for A , then the highest-priority remaining individual eligible for B , and then the highest-priority remaining individual eligible for C .

It is easy to verify that φ (which we call SSP^h in Section 4.2) respects improvements, is lower invariant, and behaves as claimed above: swapping the order of 1 and 2 has no consequence for either of these individuals, but influences the choice between 3 and 4. This behavior is ruled out by our consistency axiom.

We name this property “consistency” because it is reminiscent of the axiom of consistency used in resource allocation problems involving a varying population of individuals, which states that “An allocation rule is ‘consistent’ if the recommendation it makes for each problem ‘agrees’ with the recommendation it makes for each associated reduced problem, obtained by imagining some agents leaving with their assignments” [Thomson, 2012].

Consistency also bears some resemblance to the definition of non-bossiness given originally by Satterthwaite and Sonnenschein [1981]. That definition states that if a change to an agent's reported preference does not affect the allocation of that agent, then it should not affect any other agent's allocation. Our consistency axiom imposes an analogous requirement for changes to an agent's priority (rather than their reported preferences).

We are now ready to present our characterization of the family of outcome-based rules.

THEOREM 1.

For any $\mathcal{F} \subseteq 2^I$, the selection rule $OB_{\mathcal{F}}$ respects improvements, and is consistent and lower invariant. If φ is a selection rule that respects improvements, is consistent and is lower invariant, then there exists $\mathcal{F} \subseteq 2^I$ such that $\varphi(>) = OB_{\mathcal{F}}(>)$ for every $> \in \mathbf{P}$.

Theorem 1 implies that any selection rule either has an outcome-based description, or fails to satisfy at least one of our three axioms.

Appendix A.2 proves that our axioms are independent by providing examples of rules that fail to satisfy one axiom while satisfying the other two. For example, the over and above rule φ^{OA} as defined in Example 2 respects improvements and is lower invariant, but is not consistent. Meanwhile, optimization-based approaches that assign higher values to high-priority individuals and try to maximize total value are consistent and respect improvements, but are not lower invariant.

4 SPECIAL CASES OF INTEREST: RESERVES AND QUOTAS

Outcome-based selection rules can implement many different policies, depending on how the feasible selections $\mathcal{F}$ are defined. In this section, we consider several procedures that have been proposed in the literature and used in practice. We focus on two types of feasibility constraints, which we call *reserves* and *quotas*.

Summary of Contributions: Reserves. We introduce three ways to define reserve-based feasible selections. The first (hard reserves) requires that each position go to an eligible individual. The others (maximum and maximal reserves) allow ineligible individuals to fill positions that would otherwise go unfilled.

We then study the relationship between outcome-based rules and slot-specific priority (SSP) algorithms, which have been used in practice and studied in prior work. We answer two questions:

- (1) Which selections can be chosen by SSP algorithms?
- (2) When do SSP algorithms implement an outcome-based rule?

For the first question, Proposition 2 establishes that the selections chosen by SSP algorithms with soft reserves are exactly those that can be chosen by an outcome-based rule with maximal reserves. However, in general SSP algorithms are not outcome-based: while they always choose an element of $\mathcal{F}^{maximal}$, they may choose a priority dominated element of this set. This inspires our second question. Proposition 3 establishes conditions under which SSP algorithms replicate the behavior of outcome-based rules with feasibility defined by hard or maximum reserves.

Summary of Contributions: Quotas.

We then turn to quotas, which define lower and upper bounds on the number of selected applicants with various traits. We show that upper quotas actually suffice to reproduce the behavior of any outcome-based rule (Proposition 4). In particular, this implies that quotas are strictly more expressive than reserves.

When feasible selections are defined by quotas, even finding a feasible selection is NP-hard. Proposition 5 identifies conditions on the traits (namely, that they form a hierarchy) which guarantee that the outcome-based rule can be computed efficiently, and that the resulting selection priority dominates all other feasible selections.

4.1 Background: Matroids and Hierarchies

In essence, $OB_{\mathcal{F}}$ selects individuals greedily, so long as doing so would not violate feasibility. In general, this may come at a significant cost. For example, suppose that there are only two feasible selections: $\{1\}$, and $\mathcal{I} \setminus \{1\}$. If individual 1 is at the top of the priority order, respecting priorities requires that we select this individual, and reject all others.

Greedy algorithms are well-known to be optimal when the underlying constraints define a matroid.² We now restate this result in our setting.

DEFINITION 6. *The feasible selections $\mathcal{F}$ induce a matroid if for any $I_1, I_2 \subseteq \mathcal{I}$ such that $I_1 \subseteq S_1 \in \mathcal{F}$, $I_2 \subseteq S_2 \in \mathcal{F}$, and $|I_1| > |I_2|$, there exists $i \in I_1 \setminus I_2$ and $S_3 \in \mathcal{F}$ such that $I_2 \cup \{i\} \subseteq S_3$.³*

LEMMA 1. *If $\mathcal{F}$ induces a matroid, then for any priority order $>$, $OB_{\mathcal{F}}(>)$ priority dominates any other feasible assignment.*

We now introduce a second important definition.

DEFINITION 7. *A family of selections $\mathcal{H} \subseteq 2^{\mathcal{I}}$ forms a **hierarchy** if for every $S, S' \in \mathcal{H}$, either S and S' are disjoint ($S \cap S' = \emptyset$) or one contains the other ($S \subseteq S'$ or $S' \subseteq S$).*

This property appears in many papers under different names, and its importance is widely recognized. Biró et al. [2010] refer to it by the name “nested,” while Huang [2010], Fleiner and Kamiyama [2016], Yokoi [2017], and Gonczarowski et al. [2019] refer to a “laminar family.” Budish et al. [2013], Kamada and Kojima [2018] and Aygün and Turhan [2026] use the term hierarchy (or hierarchical), as we do.

4.2 Reserved Positions

We now study several families of feasible selections which we collectively refer to as *reserved positions*. These families are described by a bipartite graph $G = (\mathcal{I}, \mathcal{P}, E)$, where each node of the left side correspond to an individual $i \in \mathcal{I}$, each node on the right side corresponds to a position $p \in \mathcal{P}$, and there is an edge $(i, p) \in E \subseteq \mathcal{I} \times \mathcal{P}$ only if individual i is eligible for position p . We refer to $G = (\mathcal{I}, \mathcal{P}, E)$ as the *compatibility graph*. A *matching* M of G is a subset of the edges E such that every node is incident to at most one edge in M . A *maximal matching* is a matching that is not a strict subset of any other matching. A *maximum matching* is a matching that is at least as large as any other matching of G . For any matching M of G , we denote by $\mathcal{I}_M$ the set of applicants that are incident to an edge in M :

$$\mathcal{I}_M = \{i \in \mathcal{I} : \exists \text{ a position } p \in \mathcal{P} \text{ such that } (i, p) \in M\}. \quad (1)$$

In some environments, compatibility constraints must be strictly enforced. We model these situations using what we call *hard reserves*. Formally, given a compatibility graph $G = (\mathcal{I}, \mathcal{P}, E)$, the feasible selections when using hard reserves are

$$\mathcal{F}^{\text{hard}}(G) = \{S \subseteq \mathcal{I} : \exists \text{ a matching } M \text{ of } G \text{ such that } \mathcal{I}_M = S\}. \quad (2)$$

Under hard reserves, a position may go unfilled if there is no unselected individual who is eligible for it. In many applications, it may be preferable to allow someone who would otherwise not be eligible for this position to be assigned to it (Aygün and Turhan [2022] refer to this as “de-reserving”

²See https://en.wikipedia.org/wiki/Matroid#Greedy_algorithm.

³For those already familiar with matroids in other contexts, this says that the pair consisting of $\mathcal{I}$ and the downward closure of $\mathcal{F}$ (the set of all subsets of elements of $\mathcal{F}$) is a matroid.

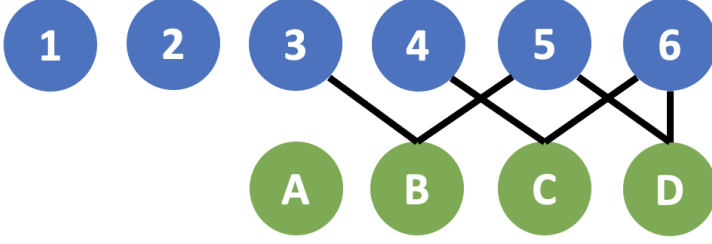


Fig. 2. An example graph G illustrating the difference between hard, soft maximum, and soft maximal reserves. Individuals are in blue and positions are in green. Hard reserves preclude matching individuals 1 and 2, and only allow three positions to fill, as nobody is eligible for position A. Maximum reserves require that three individuals from the set $\{3, 4, 5, 6\}$ are chosen, but can also select one additional applicant to fill the empty position A. Maximal reserves, meanwhile, allow the possibility of selecting the individuals $\{1, 2, 5, 6\}$, because $(5B, 6C)$ is a maximal matching in G . If low-numbered agents have higher priority, the outcome-based selection rule with feasible set $\mathcal{F}^{\text{maximal}}(G)$ will make this selection. Note that $\mathcal{F}^{\text{maximal}}(G)$ does not induce a matroid, and $\{1, 2, 5, 6\}$ does not priority dominate the feasible selection $\{1, 3, 4, 5\}$.

this position). To model such cases, we introduce two types of “soft” reserves: *maximum reserves* and *maximal reserves*. We define

$$\mathcal{F}^{\text{maximum}}(G) = \{S \subseteq \mathcal{I} : |S| \leq |\mathcal{P}| \text{ and } \exists \text{ a maximum matching } M \text{ of } G \text{ such that } \mathcal{I}_M \subseteq S\}. \quad (3)$$

$$\mathcal{F}^{\text{maximal}}(G) = \{S \subseteq \mathcal{I} : |S| \leq |\mathcal{P}| \text{ and } \exists \text{ a maximal matching } M \text{ of } G \text{ such that } \mathcal{I}_M \subseteq S\}. \quad (4)$$

Both of these definitions allow applicants to be matched to positions for which they are not eligible, so long as everyone who is eligible for that position is also selected. Both definitions also ensure that if r positions are reserved for a particular group of applicants $\mathcal{R} \subseteq \mathcal{I}$ (that is, there are r positions whose neighbors are exactly the set $\mathcal{R}$), then in every feasible selection, at least r of these applicants will be selected (or all of them, if $|\mathcal{R}| < r$).

The distinction between hard, maximum, and maximal reserves is illustrated by the example in Figure 2. For any graph G , we have $\mathcal{F}^{\text{maximum}}(G) \subseteq \mathcal{F}^{\text{maximal}}(G)$. Furthermore, any selection in $\mathcal{F}^{\text{hard}}(G)$ is also a subset of some selection in $\mathcal{F}^{\text{maximum}}(G)$. Therefore, for a given graph G , hard reserves provide the strictest interpretation of compatibility requirements. Maximum reserves grant more flexibility, and maximal reserves are the most permissive.

Both $\mathcal{F}^{\text{hard}}(G)$ and $\mathcal{F}^{\text{maximum}}(G)$ induce a matroid (Proposition 6 in Appendix B.1), implying that the selection chosen by the outcome-based rule $OB_{\mathcal{F}^{\text{maximum}}(G)}$ coincides with or priority dominates the selection from the outcome-based rule $OB_{\mathcal{F}^{\text{hard}}(G)}$, which in turn priority dominates any other selection in $\mathcal{F}^{\text{hard}}(G)$. Meanwhile, $\mathcal{F}^{\text{maximal}}(G)$ does not necessarily induce a matroid (as the example in Figure 2 demonstrates). Therefore, the selection from $OB_{\mathcal{F}^{\text{maximal}}(G)}$ may or may not priority dominate the selection from $OB_{\mathcal{F}^{\text{maximum}}(G)}$.

The remainder of this section considers the relationship between our proposed approach (using the outcome-based rule associated with $\mathcal{F}^{\text{hard}}(G)$, $\mathcal{F}^{\text{maximum}}(G)$, or $\mathcal{F}^{\text{maximal}}(G)$) and the widely-adopted “slot specific priority” (SSP) algorithms considered by Kominers and Sönmez [2016] and subsequent papers. These algorithms take as input a *precedence order* $\succ$ (ranking of positions), as well as a priority ranking $\succ_p$ for each position $p \in \mathcal{P}$. This priority ranking is over $\mathcal{I} \cup \{\emptyset\}$, where $\emptyset$ denotes the option of remaining unfilled. Given $\succ$ and $\{\succ_p\}_{p \in \mathcal{P}}$, the algorithm constructs a selection by having each position, in precedence order, take its highest-priority individual that has not yet been selected (or pass if all individuals preferred to $\emptyset$ have already been selected).

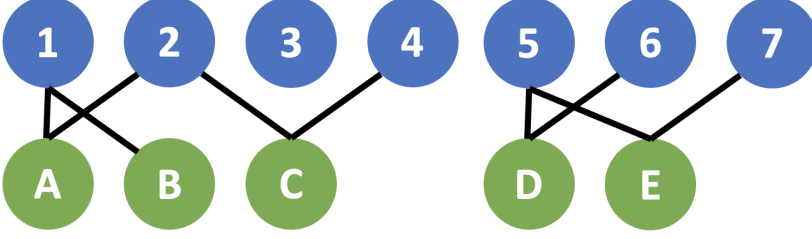


Fig. 3. An illustration of slot-specific priority algorithms with hard and soft reserves, and their limitations. There are seven applicants, with $1 > 2 > 3 > 4 > 5 > 6 > 7$ and five positions, with $A \triangleright B \triangleright C \triangleright D \triangleright E$.
 SSP_p^h : A selects 1, B passes, C selects 2, D selects 5 and E selects 7. Outcome: $\{1, 2, 5, 7\}$.
 SSP_p^s : A selects 1, B selects 2, C selects 4, D selects 5 and E selects 7. Outcome: $\{1, 2, 4, 5, 7\}$.
 $DSSP_p^s$: A selects 1, B passes, C selects 2, D selects 5 and E selects 7. Then B selects 3. Outcome: $\{1, 2, 3, 5, 7\}$.
All selections are in $\mathcal{F}^{maximal}(G)$ (Proposition 2), but the first and third are not in $\mathcal{F}^{maximum}(G)$, as any maximum matching includes $\{1, 2, 4\}$. None of these outcomes coincide with the outcome-based rule, which selects $\{1, 2, 4, 5, 6\}$ (hard or maximum reserves) or $\{1, 2, 3, 5, 6\}$ (maximal reserves).

Slot-specific priority algorithms have been used for school assignment in Boston [Dur et al., 2018] and in Chile [Correa et al., 2021], and proposed for Brazilian University admissions [Aygün and Bo, 2019]. They have also been used to allocate H1B Visas [Pathak et al., 2025] and COVID-19 vaccines [Pathak et al., 2024].

In principle, the priority orders $\succ_p$ can be arbitrary. However, in many applications, they are derived from a universal priority ordering $\succ$, and we restrict our attention to this case. There are two canonical ways to do this:

- $SSP_p^h(G, \succ)$: The neighbors of p in G , ranked according to $\succ$, followed by the outside option $\emptyset$.
- $SSP_p^s(G, \succ)$: The neighbors of p in G , ranked according to $\succ$, followed by remaining individuals ranked according to $\succ$.

We use the letters h and s to denote *hard* and *soft* reserves, to align with the terminology used by Pathak et al. [2024] and follow-up work by Evren [2025].

We also consider a second natural way to implement soft reserves, which is not a slot-specific-priority algorithm, but is closely related.

- $DSSP_p^s(G, \succ)$: First apply $SSP_p^h(G, \succ)$.

Then, fill unfilled positions with the highest priority remaining applicants.

The D in $DSSP$ stands for “delayed”, as the filling of positions by ineligible individuals is delayed until the end of the algorithm (rather than completed the first time the slot is considered). We note that the $DSSP$ rule is a special case of “dynamic reserves” from Aygün and Turhan [2020] and also of “slot-specific priorities with capacity transfers” from Avataneo and Turhan [2021].

Figure 3 illustrates the differences between the algorithms SSP_p^h , SSP_p^s , and $DSSP_p^s$. Our next result characterizes the selections found by each of these algorithms. We say that a selection S “can be selected” by an outcome-based rule $OB_{\mathcal{F}}$ if there exists a priority ordering $\succ$ such that $S = OB_{\mathcal{F}}(\succ)$. We say that S “can be selected” by an SSP procedure if there exists a precedence order $\triangleright$ and a priority ordering $\succ$ such that the SSP procedure selects S .

PROPOSITION 2 (RESERVES AND SSP ALGORITHMS). *Fix any reserve graph G .*

S can be selected by $SSP_p^h \Leftrightarrow S = I_M$ for a maximal matching M of G .

S can be selected by $OB_{\mathcal{F}^{hard}} \Leftrightarrow S = I_M$ for a maximum matching M of G .

S can be selected by $OB_{\mathcal{F}^{maximal}} \Leftrightarrow S$ can be selected by $SSP_p^s \Leftrightarrow S$ can be selected by $DSSP_p^s$

For hard reserves, this states that slot-specific priority algorithms may choose the individuals involved in any maximal matching of G , whereas our outcome-based rule $OB_{\mathcal{F}^{hard}(G)}$ always selects individuals involved in some maximum matching.

The conclusion for soft reserves establishes that the selections that can be chosen by SSP^s and $DSSP^s$ both coincide with those that can be chosen by the outcome-based rule associated with $\mathcal{F}^{maximal}(G)$.⁴ Thus, although we are the first to explicitly define the feasible set $\mathcal{F}^{maximal}(G)$, existing algorithms are implicitly selecting from it. The key difference is that slot-specific priority algorithms do not satisfy our axioms from the previous section:

- They do not respect priorities, and may choose a selection that is priority dominated (see Figure 3). Our outcome-based rule respects priorities and chooses an undominated feasible selection by Proposition 1.
- They are not consistent: improving the priority of a selected individual can change the slot that selects this individual, and thus affect who else is selected.
- The algorithms SSP^s and $DSSP^s$ are not lower invariant: changing the priority of low-priority individuals can cause a position to select a higher-ranking individual.⁵

The point about priority domination is closely related to a message from Sönmez and Yenmez [2022], who argue that slot-specific priority algorithms are flawed. Our language makes it easy to express their concerns: these algorithms may not select an element of $\mathcal{F}^{maximum}(G)$ (their Example 2), and even when they do, they might make a choice that is priority dominated (their Example 3). They propose the “meritorious horizontal choice rule,” subsequently called “smart reserves” by Pathak et al. [2024]. These algorithms are in fact equivalent to the outcome-based rule associated with $\mathcal{F}^{maximum}(G)$.

We agree with Sönmez and Yenmez [2022] that slot-specific priority algorithms are flawed. Our perspective is that policymakers should clarify which selections are considered acceptable. If the policymaker intends to use maximum reserves (as Sönmez and Yenmez [2022] assume), then we agree with their proposed solution. However, an alternative possibility is that finding a selection in $\mathcal{F}^{maximal}(G)$ is sufficient (as indicated by the widespread use of SSP^s in practice). In this case, the selection made by their meritorious horizontal choice rule may be priority dominated by another feasible selection (see Figure 3). This could be avoided by using the outcome-based rule associated with $\mathcal{F}^{maximal}(G)$, rather than $\mathcal{F}^{maximum}(G)$. One limitation is that when using $\mathcal{F}^{maximal}(G)$, computing the outcome-based rule is NP-complete (see Proposition 6 in Appendix B.1). Nevertheless, NP-complete problems have been shown to be solvable by computers in real-world selection problems [de Azevedo and Gözl, 2026, Flanigan et al., 2021a].

Evaluating the outcome-based rules $OB_{\mathcal{F}^{hard}(G)}$ and $OB_{\mathcal{F}^{maximum}(G)}$ is polynomial time, but generally requires solving a maximum matching problem. However, our next result shows that if the reserve graph has a hierarchical structure, $OB_{\mathcal{F}^{hard}(G)}$ can be implemented using $SSP^h(G, >)$

⁴It is clear that the selection chosen by $DSSP^s$ will include a maximal matching of G , as the algorithm constructs this matching in its first phase. This is less clear for SSP^s . In particular, the *matching* chosen by SSP^s might not contain *any* edge from G . (If nobody is eligible for position A , individual 1 is the only person eligible for position B , and $A \succ B$, then A could choose 1, causing B to choose 2. This matching does not contain any edge in G , but it does match all individuals (namely, individual 1) matched by a maximal matching of G .)

⁵Consider an example with 3 individuals and 2 positions. Individual 2 is eligible for position A , individual 3 is eligible for positions A and B , and position A has higher precedence than B . If the priority ordering is $1 > 2 > 3$, then SSP^s and $DSSP^s$ both select $\{2, 3\}$. However, if the ordering is $1 > 3 > 2$, then they select $\{1, 3\}$.

with a suitably chosen precedence order. To implement $OB_{\mathcal{F}^{maximum}(G)}$ under the same conditions, we cannot use $SSP_{\triangleright}^s(G, >)$, but rather must use $DSSP_{\triangleright}^s(G, >)$.⁶

PROPOSITION 3. *Suppose the sets $\{N_G(p)\}_{p \in \mathcal{P}}$ form a hierarchy. Let $\triangleright$ be any precedence order that processes more narrowly targeted positions before more general ones: $N_G(p) \subset N_G(p') \Rightarrow p \triangleright p'$. Then:*

- *The outcome-based rule $OB_{\mathcal{F}^{hard}(G)}$ is equivalent to $SSP_{\triangleright}^h(G, >)$.*
- *The outcome-based rule $OB_{\mathcal{F}^{maximum}(G)}$ is equivalent to $DSSP_{\triangleright}^s(G, >)$.*

As a special case, this implies that minimum guarantee rules are outcome-based (over-and-above rules do not satisfy the precedence condition).

4.3 Quotas

In the second family of constraints we study, there are multiple categories of applicants. For each category, there is an upper and lower limit on the number of its members that can be selected. Formally, a quota system Q is defined by

$$Q = \{(I_t, \ell_t, u_t)\}_{t \in T},$$

where T is a finite set of traits, I_t is the set of individuals with trait t , and $\ell_t, u_t \in \mathbb{N}$ are lower and upper quotas on the number of selected applicants from I_t . A selection is feasible if it satisfies quotas for every trait:

$$F(Q) = \{S \subseteq I : \ell_t \leq |S \cap I_t| \leq u_t \text{ for every } t \in T\}. \quad (5)$$

Quotas arise naturally in many contexts. For example, the US Diversity Visa Green Card lottery stipulates that at most 7% of selected applicants can be from a single country. In New York, 7% of subsidized units offered through its HousingConnect portal must be given to people with disabilities, 10% to municipal employees and veterans, and 20% to applicants from the local neighborhood. Furthermore, selected applicants are counted towards the minimum quotas for each category to which they belong (see, for example Noel, Shauna and Senat, *Emmanuella v. City of New York* [2024], Section D point 7). Panels for Citizens' Assemblies are selected to approximately match population demographics, by imposing a minimum and maximum number of people from various categories [Flanigan et al., 2021a,b]. And in Israel, the algorithm used to match students to gap year programs allows programs to specify maximum quotas and minimum targets on arbitrary sets of applicants [Gonczarowski et al., 2019].

Our first result in this section shows that maximum quotas suffice to reproduce any outcome-based selection rule.

PROPOSITION 4. *For any $\mathcal{F}$, there exists a quota system Q with all lower quotas equal to zero such that $OB_{\mathcal{F}} = OB_{F(Q)}$.*

In particular, this implies that any outcome-based rule based on the three reserve systems described in Section 4.2 can be reproduced using maximum quotas. The reverse is not true, as the following example shows.

EXAMPLE 4. *There are six applicants: two from each of three different countries. We can select at most two applicants, and at most one from each country.*

⁶To see that $SSP_{\triangleright}^s(G, >)$ will not necessarily implement $OB_{\mathcal{F}^{maximum}(G)}$, consider an example with three positions and four individuals: individual 1 is eligible for A and B, while individuals 2 and 4 are eligible for position C. The reserve graph forms a hierarchy, and the precedence order $A \triangleright B \triangleright C$ satisfies the stated condition, but $SSP_{\triangleright}^s(G, >)$ selects $\{1, 2, 4\}$, whereas $OB_{\mathcal{F}^{maximum}(G)}$ selects $\{1, 2, 3\}$ (as does $DSSP_{\triangleright}^s(G, >)$).

We show in Appendix B.2 that no reserve system can describe this feasible set.

The proof of Proposition 4 involves two steps. The first is to note that if we replace any family $\mathcal{F}$ with its downward closure (that is, we count any subset of a feasible set as feasible), this does not change the selection made by the outcome-based rule. The second step is to show that maximum quotas can be used to describe any downward-closed family. This construction involves creating a trait for each infeasible set S with the property that any proper subset of S is feasible, and creating an upper bound of $|S| - 1$ on this trait.⁷

Proposition 4 shows that upper quotas are rich enough to reproduce the behavior of *any* outcome-based rule, but the construction typically involves exponentially many traits. We show in Appendix B.2 that even when the number of traits is limited, the problem of finding a feasible selection is NP-hard in general. However, when traits are hierarchical, the family of feasible selections induces a matroid, and we can calculate the associated outcome-based rule in polynomial time.

PROPOSITION 5.

If the collection of sets $\{\mathcal{I}_t\}_{t \in T}$ forms a hierarchy, then for any upper and lower quotas $\mathbf{u}, \ell \in \mathbb{N}^T$ such that the set of feasible selections is non-empty, the feasible sets $\mathcal{F}(Q)$ induce a matroid. Therefore, the selection that respects priorities also priority dominates all other feasible selections.

If the collection of sets $\{\mathcal{I}_t\}_{t \in T}$ is not a hierarchy, then there exist upper and lower quotas $\mathbf{u}, \ell \in \mathbb{N}^T$ and a priority order $>$ such that no feasible selection priority dominates all others.

The first part (sufficiency) of Proposition 5 is proven as Lemma 3 in Yokoi [2017]. The second part (necessity) is proved in Appendix B.2.

When traits do not form a hierarchy, there may be tension between respecting priorities and other goals, such as maximizing the number of selected individuals, or minimizing the sum of their priority ranks. However, we think that outcome-based rules may nevertheless offer an appealing solution. As evidence for this, our last section considers an application where traits do not form a hierarchy (the Chilean Constitutional Assembly Election), and establishes that their chosen election procedure respects priorities.

5 ELECTION TO THE CHILEAN CONSTITUTIONAL CONVENTION

In May 2021, Chile held an election to select the members of the Constitutional Convention, which was in charge of writing a new constitution for Chile. Among its novel features was the requirement to incorporate gender balance in the convention to ensure equal participation of men and women.

To select members of the constitutional convention, each district in Chile held a separate election. In addition, there was an election to select 17 members of indigenous communities, which was conducted separately from the district elections. We focus on the district elections. In these elections, each candidate was associated with a political party, and each party was associated with a “list” (coalition of parties). We consider a simplified setting in which lists and parties coincide (all members of a list belong to the same party). For this case, the Chilean algorithm is described in Algorithm 2.

⁷To take an example, suppose that we have a total of three positions, with one reserved for a female applicant and one for a minority applicant. Then the traits constructed by our proof are as follows:

- For each pair of applicants that are neither female nor minority, impose an upper quota of 1
- For each trio of applicants without any females, impose an upper quota of 2
- For each trio of applicants without any minorities, impose an upper quota of 2
- For each set of four applicants with at least 2 women, at least 2 minorities, and at least 3 applicants who are either a woman or a minority, impose an upper quota of 3.

A	382	B	272	C	236
2	130	1	140	3	124
4	118	6	85	7	80
5	111	8	33	10	18
9	23	11	14	12	14

Fig. 4. There are three parties (A, B, C), with four candidates from each party (2 men and 2 women). Votes for each candidate are in black, and resulting priority order in green. There are 4 seats: 2 must go to men and 2 to women. Based on vote totals, $E_A = 2$, $E_B = E_C = 1$. The Chilean algorithm selects candidates $\{1, 2, 3, 9\}$. This does not priority dominate other feasible selections, such as $\{1, 2, 4, 7\}$ or $\{2, 3, 4, 6\}$. However, $\{1, 2, 3, 9\}$ is the unique outcome that respects priorities.

Algorithm 2: Simplified Chilean Constitutional Mechanism

Apportionment. Determine the number of votes earned by candidates from each party. Based on this, determine the total number of elected representatives E_i from each party i . (Roughly speaking, a party with 30% of the votes will win 30% of the seats. For details on rounding issues, see Cembrano et al. [2021].)

Initial Selection. Tentatively select the E_i highest priority candidates from each party i . Let M and W denote the total number of men and women (across all parties) selected in this step.

Swap. While the number of elected men and women differ by more than one ($|M - W| > 1$), identify the “swap-eligible” parties, which are those where a “majority gender” candidate is selected and a “minority gender” candidate is not. Among selected majority-gender candidates in swap-eligible parties, identify the one with the fewest votes, and replace them with the candidate from the same party but the opposite gender who got the most votes (among those not previously selected).

This algorithm seeks to implement appropriate party-level seat allocations while ensuring that elected representatives are as gender-balanced as possible.

The main result of this section is Theorem 2, which establishes that Algorithm 2 is equivalent to the outcome-based selection rule in which priority is determined by vote totals (candidates who received more votes have higher priority)⁸, and the feasible sets $\mathcal{F}_{chile}$ are described using quotas. In particular, we define a trait for each party and each of the genders $\{Male, Female\}$, and define $\mathcal{F}_{chile}$ to be the set of selections in which (i) exactly E_i candidates are elected from each party i , and (ii) the number of elected candidates from each gender is at least $\lfloor N/2 \rfloor$ and at most $\lceil N/2 \rceil$ (where N is the number of seats in the district).

THEOREM 2. *The selection made by Algorithm 2 and $OB_{\mathcal{F}_{chile}}(>)$ coincide.*

Theorem 2 implies that Algorithm 2 finds the outcome-based selection efficiently even though the traits that define $\mathcal{F}_{chile}$ do not form a hierarchy (the sets of candidates of a particular gender and candidates from a particular party overlap, but neither contains the other). This is possible because calculating the outcome-based rule associated with $\mathcal{F}_{chile}$ is a special case of the problem of finding a common base of two matroids [Schrijver, 2003, Chapter 41], for which efficient algorithms are available. Specifically, if the feasible selections are the common bases of two matroids, the selection that respects priorities can be found in polynomial-time using standard algorithms for the matroid intersection problem, see Edmonds [1970] or Lawler [1975]⁹.

By Proposition 5, we cannot expect a feasible selection that priority-dominates all others. This is illustrated in Figure 4. In that example, the Chilean algorithm selects candidate 9, who received only 23 votes, while other feasible selections (such as $\{2, 3, 4, 6\}$) do not take such a low-ranking

⁸We assume that no two candidates receive the same number of votes, so that $>$ is a strict ordering. We are uncertain about how the Chilean algorithm is defined on pathological examples where, say, all candidates receive the same number of votes.

⁹Indeed, the two matroids in the Chilean context—one for the party and one for the gender—are both laminar, meaning the feasible collection forms a bihierarchy. This special case can be thought of as a linear programming problem with a totally unimodular constraint matrix as highlighted by Edmonds [1970]; see also [Budish et al., 2013].

candidate. Unsuccessful candidates with more than 23 votes might demand an explanation. Theorem 2 provides a new way to explain this outcome: any unsuccessful candidate could only have been selected by excluding a candidate with more votes or by violating gender/party constraints. This fact is not obvious from algorithmic descriptions of the Chilean algorithm.¹⁰ We take the fact that Chile chose this rule – despite its potential to elect candidates with very low vote totals – as suggestive evidence that even if the feasible selections do not induce a matroid, outcome-based rules may be appealing in practice.

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¹⁰See for example the descriptions at <https://www.biobiochile.cl/especial/una-constitucion-para-chile/noticias/2021/02/26/como-funcionara-la-paridad-de-genero-en-la-eleccion-de-constituyentes.shtml> or https://www.cnnchile.com/eleccion2021/paridad-ajuste-genero-eleccion-constituyentes_20210324/

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A PROOFS FROM SECTION 3

A.1 Proof of Theorem 1

We divide the proof into two parts: Lemma 2 establishes that for any collection of feasible selections $\mathcal{F} \subseteq 2^I$, the rule $OB_{\mathcal{F}}$ respects improvements, is consistent and is lower invariant. Lemma 3 shows that if a selection rule satisfies these three properties, then it must be equivalent to $OB_{\mathcal{F}}$ for some suitably chosen $\mathcal{F}$.

LEMMA 2. *For any collection $\mathcal{F} \subseteq 2^I$, the selection rule $OB_{\mathcal{F}}$ respects improvements, and is consistent and lower invariant.*

PROOF. It is clear from the definition of $OB_{\mathcal{F}}$ that it respects improvements and is lower invariant. Consistency needs only to be shown for transpositions of adjacent agents in the priority order (as all changes to the position of individual i can be expressed as a sequence of transpositions). Suppose that i and j appear consecutively in the priority order, and let S be the set of applicants selected before these individuals are considered. We analyze the following cases.

- It is feasible (given the prior commitments to agents in S) to select both i and j .
In this case, swapping positions has no effect for either agent: both will be selected.
- It is feasible to select either i or j , but not both.
Swapping positions changes the outcome for both i and j .
- It is feasible to select i , but infeasible to select j .
Swapping positions has no effect for either agent: i will be selected but j will not.
- It is feasible to select j , but infeasible to select i .
Swapping positions has no effect for either agent: j will be selected but i will not.
- It is infeasible to select either i or j .
Swapping positions has no effect for either agent: neither will be selected.

Now suppose that the swap does not affect agent i . By the case analysis above, it also must not affect agent j . Therefore, after $OB_{\mathcal{F}}$ has processed i and j , it has made the same set of commitments regardless of the order in which they appear. By the definition of $OB_{\mathcal{F}}$, this implies that the order in which i and j appear does not affect the decisions made for all subsequent agents, so $OB_{\mathcal{F}}$ is consistent. $\square$

LEMMA 3. *Let φ be a selection rule that respects improvements, and is consistent and lower invariant. Then there exists $\mathcal{F} \subseteq 2^I$ such that*

$$\varphi(>) = OB_{\mathcal{F}}(>) \text{ for any } > \in \mathbf{P}. \quad (6)$$

PROOF. Define $\mathcal{F} = \{\varphi(>)\}_{> \in \mathbf{P}}$ to be the image of φ . Fix a priority ordering $>$, and without loss of generality, label agents so that agent 1 has highest priority, followed by 2, etc. Fix some k , and let $S = \{1, \dots, k-1\} \cap \varphi(>)$ be the subset of the first $k-1$ agents who are selected by φ . To prove the result, it suffices to show that if φ and $OB_{\mathcal{F}}$ make the same decision for agents 1, 2, ..., $k-1$, then they also make the same decision for agent k .

We first show that $k \in \varphi(>)$ implies $k \in OB_{\mathcal{F}}(>)$. If k is selected by φ , then $\varphi(>)$ is a feasible selection containing $\{k\} \cup S$, and therefore $OB_{\mathcal{F}}$ will select k .

The more interesting direction is showing that $k \in OB_{\mathcal{F}}(>)$ implies $k \in \varphi(>)$. By definition of $\mathcal{F}$, if $k \in OB_{\mathcal{F}}(>)$ then there must exist a priority ordering $>_0$ such that $\{k\} \cup S \subseteq \varphi(>_0)$. We will construct a sequence of priority orderings $\{>_i\}_{i=1}^k$ as follows: $>_i$ is constructed from $>_{i-1}$ by taking the i^{th} ranked individual (according to $>$), and moving that individual into position i . We claim that for $1 \leq i \leq k$, we have the following:

- A) $>_i$ and $>$ produce the same outcome for individuals 1, 2, ..., $k-1$.

B) Individual k is selected when the priority order is $>_i$.

These clearly hold for $i = 0$: by definition, $\varphi(>_0)$ contains S , and if it contained any other agent in $\{1, \dots, k-1\}$, then $OB_{\mathcal{F}}(>)$ would also include this agent, contradicting that φ and $OB_{\mathcal{F}}$ agree on the first $k-1$ individuals.

Now suppose that these assumptions hold for $i < k-1$. Note that:

- I. $>_{i+1}$ produces the same outcome as $>$ for agent $i+1$ (the first $i+1$ positions of $>_i$ and $>$ are identical by construction, so this follows from lower invariance of φ).
- II. $>_i$ and $>_{i+1}$ produce the same outcome for *all* agents (by step I. and inductive assumption A), they produce the same outcome for $i+1$, and φ is consistent).

In particular, this implies that the inductive assumptions hold for $i+1$.

This inductive logic implies that agent $k \in \varphi(>_{k-1})$. Therefore, when we construct $>_k$ from $>_{k-1}$, k must continue to be selected (because φ respects improvements). But then lower invariance implies that $k \in \varphi(>)$, completing the proof. $\square$

A.2 Examples Demonstrating Independence of Our Three Axioms

EXAMPLE 5 (RESPECTS IMPROVEMENTS, LOWER-INVARIANT BUT NOT CONSISTENT). *Consider a setting where there are multiple minority groups that are disjoint, that is, each applicant can be a member of at most one minority. There are two types of seats: (i) unreserved seats that can be allocated to any applicant, and (ii) for each minority, there are reserved seats that can be allocated only to members of the minority. Let φ be the selection rule that, for any priority order $>$, generates a selection according to the following procedure:*

- (1) *Unreserved seats are allocated based on the priority order $>$.*
- (2) *For each of the (mutually exclusive) minorities, positions that are set aside are allocated to the remaining members of the minority again based on the priority order $>$.*

We start by showing that no selection rule $OB_{\mathcal{F}}$ can replicate the behavior of φ . Consider an instance with three applicants $\mathcal{I} = \{1, 2, 3\}$, one open seat and one seat reserved for members of the minority group $M = \{2, 3\}$. Let

$$1 > 2 > 3 \text{ and } 2 >' 1 >' 3.$$

From the definition of φ , we must have that $OB_{\mathcal{F}}(>) = \{1, 2\}$ so $\{1, 2\} \in \mathcal{F}$. In addition, it should be that $OB_{\mathcal{F}}(>') = \{2, 3\}$. However, selection $\{2, 3\}$ does not respect priorities $>'$ as $1 >' 3$ and $\{1, 2\} \in \mathcal{F}$.

From the instance above, we can verify that φ is not consistent. Notice that under both orders (i) applicant 2 is selected, and (ii) applicant 1 is preferred to applicant 3. However, the overall selections differ.

EXAMPLE 6 (RESPECTS IMPROVEMENTS, CONSISTENT BUT NOT LOWER-INVARIANT). *Consider an instance with three applicants $\mathcal{I} = \{1, 2, 3\}$. We define the selection rule φ as follows:*

- (1) *If 3 is not the worst priority applicant, then select applicant 3.*
- (2) *Otherwise, select applicants 1 and 2.*

We begin by showing that no selection rule $OB_{\mathcal{F}}$ can replicate the behavior of φ . Let

$$1 > 2 > 3 \text{ and } 1 >' 3 >' 2.$$

By (2) have that $OB_{\mathcal{F}}(>) = \{1, 2\}$ so $\{1, 2\} \in \mathcal{F}$. In addition, by (1) we have that $OB_{\mathcal{F}}(>') = \{3\}$. However, selection $\{3\}$ does not respect priorities $>'$ as $1 >' 3$ and $\{1, 2\} \in \mathcal{F}$.

It is easy to see that φ is consistent. Notice that the two feasible selections are disjoint. Hence, if the outcome of an applicant remains unchanged, then the overall selection must be the same.

We now show that φ respects improvements. On one hand, if the priority of applicant 3 is improved, then she will be selected and weakly better. On the other hand, applicants $\{1, 2\}$ are only selected when they are in the first two positions. In this situation, the only improvement feasible is to increase the priority of the applicant in the second position. Of course, this maintains the overall selection and this agent is weakly better.

Finally, we see that φ is not lower-invariant. Notice that applicant 1 has the top priority in $>$ and $>'$. However, φ only selects applicant 1 under $>$.

EXAMPLE 7 (CONSISTENT, LOWER-INVARIANT BUT DOES NOT RESPECT IMPROVEMENTS). Consider an instance with two applicants $I = \{1, 2\}$. Let the selection rule φ be defined as follows:

- (1) If applicant 1 has the top priority, select all applicants.
- (2) Otherwise, select nobody.

We start by showing that no selection rule $OB_{\mathcal{F}}$ can replicate the behavior of φ . Let

$$1 > 2 \text{ and } 2 >' 1.$$

From (1) above, we must have that $OB_{\mathcal{F}}(>) = \{1, 2\}$ so $\{1, 2\} \in \mathcal{F}$. In addition, it should be that $OB_{\mathcal{F}}(>') = \emptyset$. Of course, selecting no applicants does not respect priorities $>'$ as applicant 2 has the first priority and $\{1, 2\} \in \mathcal{F}$.

From the instance above, we see that φ does not respect improvements. Notice that the priority of applicant 2 is better in $>'$ than in $>$ but it is only selected under $>$.

Clearly, φ is consistent. Note that φ either selects all applicants or nobody. Hence, if the status of an applicant remains unchanged so does the overall selection.

Finally, we show that φ is lower-invariant. This follows from the fact that the selection is determined by the applicant who has the top priority.

B PROOFS FROM SECTION 4

PROOF OF LEMMA 1. Lemma 1 is now standard and goes back to Gale [1968]. We include a proof for the sake of completeness.

For the sake of contradiction, suppose that there exists a priority order $>$ and a feasible selection S , such that S is not priority dominated by $OB_{\mathcal{F}}(>)$.

For each selection S and integer $k \in \{1, \dots, |S|\}$, we let $S^k \subseteq S$ be the subset containing the k highest priority individuals in S , that is,

$$S_k = \{i_1^S, \dots, i_k^S\}.$$

Let $k^* \in \{1, \dots, |S|\}$ be the highest-priority index for which

$$i_{k^*}^S > i_{k^*}^{OB_{\mathcal{F}}(>)}.$$

The existence of k^* is guaranteed as S is not dominated by $OB_{\mathcal{F}}(>)$. Consider $OB_{\mathcal{F}}(>)^{k^*-1}$ and S^{k^*} . Because $\mathcal{F}$ induces a matroid, there must exist $j \in S^{k^*} \setminus OB_{\mathcal{F}}(>)^{k^*-1}$ and $S' \in \mathcal{F}$ such that $OB_{\mathcal{F}}(>)^{k^*-1} \cup \{j\} \subset S'$. Then $OB_{\mathcal{F}}(>)$ does not respect priorities as $j \notin OB_{\mathcal{F}}(>)$ but can be selected without displacing any higher priority individual in $OB_{\mathcal{F}}(>)$. □

B.1 Reserved Positions

PROOF OF PROPOSITION 2. We start with SSP^h . Suppose S is the set of individuals chosen by SSP^h for some precedence order $\triangleright$ and priority ordering $>$. A position p remains unfilled only when all of its compatible individuals are already chosen when p is processed by SSP^h . Thus, the

matching chosen by SSP^h is a maximal matching, as no unfilled position can be added. For the converse, suppose S is the set of individuals matched by some maximal matching M of G . Start with an arbitrary order of the matched edges; and pick $(\succ, \triangleright)$ in which the individuals and positions are ordered according to the chosen ordering of the matched edges, followed by an arbitrary ordering of the unmatched individuals and unfilled positions respectively. It is straightforward to check that SSP^h with these orderings selects S .

We now turn to the outcome-based rule associated with hard reserves. It is clear that for any maximum matching M of G , the individuals $\mathcal{I}_M$ could be chosen by the outcome-based selection rule: simply place them at the top of the priority list, in any order. To establish the converse (that any selection produced by the outcome-based rule corresponds to a maximum matching of G), we use the fact that $\mathcal{F}^{hard}(G)$ induces a matroid. By definition, the outcome-based rule selects S only if $S = \mathcal{I}_M$ for some matching M of G . Suppose that $|M| < |M'|$, where M' is any maximum matching of G . Then the matroid property implies that there is an individual $i \in \mathcal{I}_{M'} \setminus \mathcal{I}_M$ such that $\mathcal{I}_M \cup \{i\}$ is feasible (can be matched by a matching M''). It follows that $\mathcal{I}_M$ cannot be chosen by the outcome-based rule, as when this rule processed individual i , it would have accepted. Therefore, the outcome-based rule must select the individuals matched by a maximum matching of G .

We now turn to soft reserves. It is clear that the outcome-based rule associated with $\mathcal{F}^{maximal}(G)$ can select any maximal element of $\mathcal{F}^{maximal}(G)$. We show that these are exactly the selections that can be chosen by $DSSP^s$ and SSP^s . We first show that the outcome of these algorithms must be a maximal element of $\mathcal{F}^{maximal}(G)$.

Recall that $DSSP^s$ first applies SSP^h (phase 1), and then offers the unfilled positions to the highest priority remaining applicants (phase 2). By the result for SSP^h that we just proved, the outcome at the end of the first phase is a maximal matching; in the second phase, additional individuals are selected until either all positions are filled or all individuals are selected. Thus, the final selection contains a maximal matching, and is of maximum cardinality, making it a maximal element in $\mathcal{F}^{maximal}(G)$.

We establish the corresponding result for SSP^s using an induction argument. Let G_k be the subgraph of G containing the first k positions (according to $\triangleright$), and let I_k be the individuals selected by the algorithm after k steps. The induction hypothesis is that there is a maximal matching M_k of G_k such that the individuals matched in M_k are a subset of I_k . This is trivially true for the base case when $k = 0$. Suppose the position examined at step $k + 1$ of the algorithm is p and the individual selected is i (if no individual is selected, then we are in the trivial case where selecting all of $\mathcal{I}$ is feasible). We need to prove that there is a maximal matching M_{k+1} of G_{k+1} such that all the matched individuals are a subset of $I_k \cup \{i\}$. By the induction hypothesis, there is a maximal matching M_k of G_k such that all the matched individuals are a subset of I_k . If p and i are compatible, then set $M_{k+1} = M_k \cup \{(i, p)\}$, which is naturally a maximal matching in which all the matched individuals are a subset of $I_k \cup \{i\}$. If p and i are not compatible, we consider two cases:

- There is an individual $i' \in I_k$ who is not matched by M_k and is compatible with p .
- There is no such individual.

In the first case, set $M_{k+1} = M_k \cup (i', p)$. Then M_{k+1} is maximal in G_{k+1} , as we have no “new” unfilled positions (relative to M_k). In the second case, we claim that M_k is maximal in G_{k+1} , and therefore that we can set $M_{k+1} = M_k$. This is true because (by definition) all individuals compatible with p are matched by M_k . This completes the proof by induction. The final selection therefore contains a maximal matching of G . Moreover, at the end of the algorithm, either all positions are filled or all individuals are selected. Thus, the final selection contains a maximal matching, and is of maximum cardinality, making it a maximal element in $\mathcal{F}^{maximal}(G)$.

We now establish the converse: given any maximal element $S \in \mathcal{F}^{\text{maximal}}(G)$, there is some priority ranking $>$ and some precedence order $\triangleright$ for which the outcomes of $DSSP^s$ and SSP^s are S . While the statement allows us to choose different orderings for these two algorithms, we construct a single $(>, \triangleright)$ that works for both. Because $S \in \mathcal{F}^{\text{maximal}}(G)$, there is a maximal matching M of G such that all the individuals matched by M are included in S . Fix any precedence order $\triangleright$ in which the positions matched by M appear before unmatched positions. The priority ranking $>$ is constructed in three tiers as follows: (tier 1) all the individuals matched in M , with their ordering dictated by the ordering of the positions they are matched to: for any $(i, p), (i', p') \in M$, $p \triangleright p' \implies i > i'$; (tier 2) all the individuals in S who are not matched in M , ordered arbitrarily; (tier 3) all remaining individuals, ordered arbitrarily. Both SSP^s and $DSSP^s$ will process the positions matched by M and will match them to their partners in M ; once these positions are cleared, the unmatched positions are evaluated. Because M is a maximal matching, none of the remaining positions is compatible with any of the remaining individuals. Thus, SSP^s and $DSSP^s$ operate in exactly the same manner, assigning all the tier 2 individuals to positions, yielding exactly the selection S . $\square$

Note that for this equivalence to hold, we need the flexibility to choose *both* the priority and the precedence order. For a given precedence order $\triangleright$, not all elements of $\mathcal{F}^{\text{maximal}}(G)$ are achievable. For example, suppose that there are three individuals and two positions, with individual 2 eligible for positions A and B and individual 3 eligible for position B . Then $\{1, 2\} \in \mathcal{F}^{\text{maximal}}(G)$, but a slot-specific priority algorithm with $A \triangleright B$ will always choose the selection $\{2, 3\}$, regardless of the priority order.

PROOF OF PROPOSITION 3. (Part 1: Equivalence under Hard Reserves)

We are given a graph G such that the sets $\{N_G(p)\}_{p \in \mathcal{P}}$ form a hierarchy, and a precedence order $\triangleright$ such that $N_G(p) \subset N_G(p')$ implies $p \triangleright p'$. We first prove that the outcome-based rule on hard-reserves and the SSP^h find the same outcome. Our proof is by induction on the number of slots.

Base Case. If there is only one slot, the precedence order $\triangleright$ is trivial. The SSP^h mechanism evaluates the single slot $p \in P$ and selects the highest-priority individual compatible with the slot. The outcome-based rule evaluates individuals in decreasing order of $>$ and selects the first individual i such that $i \in N(p)$. Both mechanisms select the identical individual, establishing the base case.

Inductive Step. Suppose there is more than one slot. Let p^* be the first slot processed under the precedence order $\triangleright$. From the condition on the precedence order, either $N(p^*) \subseteq N(p)$ or $N(p^*) \cap N(p) = \emptyset$ for any other slot p .

If no individual is compatible with p^* , p^* must be unassigned and we can omit it from the problem, so the result follows by induction. Otherwise, p^* is assigned to individual i^* , the highest priority individual eligible for slot p^* .

Observe that, for the given priority ordering, the outcome-based algorithm $\mathcal{F}^{\text{hard}}(G)$ must select i^* : when individual i^* is processed, slot p^* must be unassigned, as no higher priority individual is eligible for p^* .

Consider the reduced problem G^* in which the individual i^* and the slot p^* are removed. We next show that $S \cup \{i^*\} \in \mathcal{F}^{\text{hard}}(G)$ if and only if $S \in \mathcal{F}^{\text{hard}}(G^*)$. One direction is clear: if $S \in \mathcal{F}^{\text{hard}}(G^*)$, then we can assign slot p^* to i^* , which proves $S \cup \{i^*\} \in \mathcal{F}^{\text{hard}}(G)$. Now suppose $S \cup \{i^*\} \in \mathcal{F}^{\text{hard}}(G)$. If p^* is assigned to i^* or if p^* is unassigned, the result follows immediately. Otherwise, p^* is assigned to some $i \neq i^*$ and some slot $p \neq p^*$ is assigned to i^* . By our assumption on the precedence order, i is eligible for p , so we can interchange the assignments of p and p^* to get a valid matching in which p^* is assigned to i^* .

We have just established that, conditional on selecting i^* , the feasible continuations in G are exactly the feasible selections in G^* , from which it follows that the selection of $OB_{\mathcal{F}^{hard}(G)} = OB_{\mathcal{F}^{hard}(G^*)} \cup \{i^*\}$. By the induction hypothesis, $OB_{\mathcal{F}^{hard}(G^*)}$ and $SSP_{\triangleright}^h(G^*)$ select the same individuals. Finally, note that $SSP_{\triangleright}^h(G, \succ) = SSP_{\triangleright}^h(G^*, \succ) \cup \{i^*\}$. This completes the inductive argument and establishes the result for hard reserves.

(Part 2: Equivalence under Soft Reserves) We now turn to the soft maximum reserve rule. Recall that the delayed algorithm $DSSP_{\triangleright}^s(G, \succ)$ operates in two distinct phases. In Phase 1, it executes the hard algorithm $SSP_{\triangleright}^h(G, \succ)$. By the earlier argument for hard reserves, this phase constructs a maximum-cardinality matching B over the compatibility graph G that is the unique priority-dominant outcome. In Phase 2, the algorithm identifies any positions left unfilled at the end of Phase 1 and populates them using the highest-priority applicants remaining in $\mathcal{I} \setminus B$, regardless of their compatibility, until the positions are filled or the applicant pool is entirely exhausted.

To establish equivalence with the axiomatic rule, we exploit the combinatorial structure of the underlying matching matroid. Let $\mathcal{M}$ denote the standard matching matroid defined on the ground set of individuals $\mathcal{I}$, where a subset of individuals is independent if and only if it can be simultaneously matched to slots in $\mathcal{P}$ under the compatibility graph G . By definition, the bases of $\mathcal{M}$ correspond exactly to the maximum-cardinality matchings in G . Under the soft maximum outcome-based reserve rule, a selection $S \subseteq \mathcal{I}$ can be chosen if and only if it has a total cardinality of $\min(|\mathcal{I}|, |\mathcal{P}|)$ and fully contains a maximum-cardinality matching—that is, S must contain some basis B' of $\mathcal{M}$.

A foundational property of matroid optimization is that the global priority-maximizing independent set can be constructed via an element-by-element greedy choice Gale [1968]. Thus, maximizing a strictly ordered priority function $\succ$ over a capacitated selection that must contain a basis can be perfectly decoupled into a two-stage optimization problem: (i) Identify the unique, priority-maximizing basis B of the underlying matching matroid $\mathcal{M}$; and (ii) Fill the remaining available capacity up to $\min(|\mathcal{I}|, |\mathcal{P}|)$ by greedily selecting the absolute highest-priority individuals from the remaining available pool $\mathcal{I} \setminus B$.

By our argument for hard reserves, Phase 1 of the $DSSP^s$ precisely resolves Stage 1 by identifying this priority-maximal maximum matching B . Phase 2 subsequently resolves Stage 2 by expanding B with the absolute highest-priority remaining individuals in $\mathcal{I} \setminus B$.

Because the greedy choice property ensures that this two-step sequential extension yields a final selection that globally priority-dominates all other feasible alternatives containing a maximum matching, the resulting allocation coincides with the unique, axiomatic outcome-based selection rule $OB_{\mathcal{F}^{maximum}(G)}(\succ)$. $\square$

PROPOSITION 6.

For any graph G , $\mathcal{F}^{hard}(G)$ and $\mathcal{F}^{maximum}(G)$ induce a matroid, and the rules $OB_{\mathcal{F}^{hard}(G)}$ and $OB_{\mathcal{F}^{maximum}(G)}$ can be evaluated in polynomial time.

The family $\mathcal{F}^{maximal}(G)$ may not induce a matroid, and the problem of evaluating $OB_{\mathcal{F}^{maximal}(G)}$ is NP-complete.

PROOF OF PROPOSITION 6. We start with maximal reserves. The example in Figure 2 shows that the collection $\mathcal{F}^{maximal}(G)$ need not induce a matroid: $\{1, 2, 5\}$ is a subset of the feasible selection $\{1, 2, 5, 6\}$, and $\{1, 3, 4, 5\}$ is feasible, but neither 3 nor 4 can be added to $\{1, 2, 5\}$.

To establish that evaluating the associated outcome-based rule is NP-complete, we give a reduction from the problem “minimum maximal matching” (MMM) of deciding whether a bipartite graph G admits a maximal matching of size at most k . This problem was shown to be NP-complete by Yannakakis and Gavril [1980]. Consider an instance of MMM, that is, a bipartite graph $G = (U, V, E)$

and a positive integer k . We generate an instance $I(G)$ of the selection problem under maximal reserves as follows. We set $\mathcal{P} = V$ and $I = I^1 \cup I^2$, where $I^1 = U$ and I^2 consists of $|V|$ individuals who are not connected to any position. Let the priority order $>$ place all individuals in I^2 ahead of all individuals in I^1 . Then G admits a maximal matching of size at most k if and only if there is a selection in $\mathcal{F}^{\text{maximal}}(G)$ containing the first $|V| - k$ individuals from $>$.

We now prove the positive result for hard reserves. They induce the well-known “matching matroid” or “transversal matroid” [Oxley, 2011, Section 1.6]. This implies that if we assign positive weights to individuals such that higher-priority individuals get higher weights, and then solve for a maximum weight matching of individuals to positions, the resulting selection priority dominates any other feasible selection (this would not be true for feasible sets that do not induce a matroid). Furthermore, this maximum weight matching problem can be solved in polynomial time, for example by using the Hungarian algorithm.

We now turn to maximum reserves. We assume that $|\mathcal{P}| \leq |I|$; the case where $|\mathcal{P}| > |I|$ is trivial, as in that case the outcome-based rule will always select all individuals. Then maximum reserves induce what is known as the elongation of the matroid induced by hard reserves to rank $|\mathcal{P}|$ (Welsh, 1976; Oxley, 2011, Exercise 1.3.7). This is easiest to explain using the basis definition of a matroid. Given any matroid $(I, \mathcal{F})$, let $\mathcal{B}$ be the maximal elements of $\mathcal{F}$. These are called the *bases* of the matroid, and the definition of a matroid ensures that they all have the same size, which we will call m . The bases can also be used to define the matroid, by simply taking $\mathcal{F}$ to be all subsets of elements of $\mathcal{B}$. For any integer $k \geq m$, define

$$\mathcal{B}^k = \{S \subseteq I : |S| = k \text{ and } S \supseteq B \text{ for some } B \in \mathcal{B}\}.$$

Then $(I, \mathcal{B}^k)$ is itself a matroid, called the elongation of $(I, \mathcal{B})$ to rank k . In our context, $\mathcal{F} = \mathcal{F}^{\text{hard}}(G)$, the bases $\mathcal{B}$ are the sets of individuals matched by a maximum matching of G , $k = |\mathcal{P}|$, and the sets $\mathcal{B}^k$ are the maximal elements of $\mathcal{F}^{\text{maximum}}(G)$.

To evaluate the outcome-based rule associated with $\mathcal{F}^{\text{maximum}}(G)$ in polynomial time, we first evaluate the outcome-based rule associated with $\mathcal{F}^{\text{hard}}(G)$. This gives us a set of individuals S^* which priority dominates any other set from $\mathcal{F}^{\text{hard}}(G)$. We then accept the $|\mathcal{P}| - |S^*|$ remaining individuals with the highest priority. This clearly gives us a selection in $\mathcal{F}^{\text{maximum}}(G)$, as the individuals S^* are all matched by a maximum matching of G . Let R^* be the set of individuals accepted in the second stage. We claim that $I^* = S^* \cup R^*$ priority dominates any other selection in $\mathcal{F}^{\text{maximum}}(G)$, from which it follows that this must be the selection chosen by the associated outcome-based rule.

To establish this claim, let $I \in \mathcal{F}^{\text{maximum}}(G)$, fix an integer $k \leq |I|$, and let I_k denote the k highest priority individuals. We must show that $|I \cap I_k| \leq |I^* \cap I_k|$. Because $I \in \mathcal{F}^{\text{maximum}}(G)$, we can write $I = S \cup R$, where S consists of a set of individuals involved in a maximum matching of G and $|R| \leq \ell$, where $\ell = |\mathcal{P}| - |S| = |\mathcal{P}| - |S^*|$ is the number of positions left unfilled by every maximum matching. Then

$$\begin{aligned} |I \cap I_k| &\leq \min\{\ell + |S \cap I_k|, k\} \\ &\leq \min\{\ell + |S^* \cap I_k|, k\} \\ &= |I^* \cap I_k|. \end{aligned}$$

The first inequality follows because $|R| \leq \ell$, and thus the number of top k individuals in I cannot exceed ℓ plus the number in S (and of course also cannot exceed k). The second inequality follows because S^* priority dominates S (as S^* is the result of evaluating the outcome-based rule for $\mathcal{F}^{\text{hard}}(G)$, which induces a matroid). The final equality follows because $R^* = I^* \setminus S^*$ consists of the ℓ

highest-priority individuals that are not in S^* . This completes the proof that I^* priority dominates any other $I \in \mathcal{F}^{\text{maximum}}(G)$, and therefore is the selection chosen by $OB_{\mathcal{F}^{\text{maximum}}(G)}$. $\square$

B.2 Quotas

Because the behavior of an outcome-based rule with feasible selections $\mathcal{F}$ is equivalent to its behavior on the downward closure of $\mathcal{F}$, Proposition 4 follows immediately from the following.

LEMMA 4 (QUOTA REPRESENTATION OF DOWNWARD-CLOSED FAMILIES). *Let $\mathcal{F} \subseteq 2^I$ be non-empty and downward closed family feasible selections: if $S \subseteq S' \subseteq I$ and $S' \in \mathcal{F}$, then $S \in \mathcal{F}$. Then there is a quota system Q involving only maximum quotas such that $F(Q) = \mathcal{F}$.*

PROOF. Let $\mathcal{B}$ be the collection of inclusion-minimal infeasible sets:

$$\mathcal{B} = \{B \subseteq I : B \notin \mathcal{F} \text{ and } B' \in \mathcal{F} \text{ for every } B' \subset B\}.$$

For each $B \in \mathcal{B}$, introduce a trait t_B with

$$\mathcal{I}_{t_B} = B, \quad \ell_{t_B} = 0, \quad u_{t_B} = |B| - 1.$$

Let Q denote the resulting quota system. Since $\mathcal{F}$ is nonempty and downward closed, $\emptyset \in \mathcal{F}$, so every $B \in \mathcal{B}$ is nonempty and $u_{t_B} \geq 0$.

We show that $F(Q) = \mathcal{F}$. First suppose $S \in \mathcal{F}$. If S violated some quota, then for some $B \in \mathcal{B}$ we would have

$$|S \cap B| > |B| - 1.$$

Since $S \cap B \subseteq B$, this implies $B \subseteq S$. But $B \notin \mathcal{F}$, while $S \in \mathcal{F}$, contradicting downward closure. Therefore $S \in F(Q)$.

Conversely, suppose $S \notin \mathcal{F}$. Since I is finite, there exists some $B \in \mathcal{B}$ such that $B \subseteq S$. For the corresponding trait t_B ,

$$|S \cap \mathcal{I}_{t_B}| = |S \cap B| = |B| > |B| - 1 = u_{t_B}.$$

Hence $S \notin F(Q)$. $\square$

LEMMA 5. *The feasible family in Example 4 cannot be described by a reserve system (hard, maximum, or maximal).*

PROOF. Recall that in this example, there are six applicants, with two from each of three different countries. We can select at most one applicant from each country, and at most two overall. Let the six applicants be $a_1, a_2, b_1, b_2, c_1, c_2$, where a_1 and a_2 are from country A , b_1 and b_2 from country B and c_1, c_2 from country C .

First consider hard reserves. If two applicants are individually feasible but cannot be selected together, they must both be eligible for the same position (and only this position). Therefore a_1 and a_2 must be eligible for a single position p_a . Similarly, b_1 and b_2 must be eligible for position p_b , and c_1 and c_2 for position p_c . If these three positions are all distinct, then we could select three applicants, which is not allowed. However, if two of these positions coincide (say $p_a = p_b$) then it is not possible to select an applicant from country A alongside one from country B .

Meanwhile, for soft reserves, if all maximal (or maximum) matchings match all positions, then the feasible selections are exactly the maximal feasible selections under hard reserves. If there are maximal (or maximum) matchings that do not match all positions, then no constraints are placed on the individuals chosen to fill the remaining positions, so there is a possibility of selecting two applicants from the same country. $\square$

PROOF OF PROPOSITION 5. The sufficiency (first) part of the result is a restatement of Lemma 3 in Yokoi [2017]. To show necessity, suppose that $\mathcal{I}_T$ does not form a hierarchy. Then there exist traits t_1 and t_2 and individuals i_1, i_2, i_3 such that $i_1 \in \mathcal{I}_{t_1} \setminus \mathcal{I}_{t_2}$, $i_2 \in \mathcal{I}_{t_2} \setminus \mathcal{I}_{t_1}$, and $i_3 \in \mathcal{I}_{t_1} \cap \mathcal{I}_{t_2}$.

Set all lower quotas to be zero, set an upper quota of 1 for each of traits t_1 and t_2 , and a (vacuous) upper quota of $|\mathcal{I}|$ for all other traits. Let $3 > 2 > 1$ be the three highest-ranked individuals according to $>$. Then the outcome-based rule selects individual 3, along with everyone who does not have traits t_1 or t_2 . However, it is also feasible to select individuals 1 and 2 along with everyone who does not have traits t_1 or t_2 . This selection is not priority dominated by the selection of the outcome-based rule. By Proposition 1, it follows that there is no feasible selection that priority dominates all others. $\square$

We now show that determining whether there exists a feasible selection is NP-complete.

PROPOSITION 7. *Determining whether $F(Q)$ defined in (5) is empty is NP-complete.*

This follows from Lemma 6 and/or Lemma 7, each of which proves the result for a special case of the general quotas problem.

LEMMA 6. *Suppose there is a trait $0 \in T$, such that $\mathcal{I}_0 = \mathcal{I}$ and $\ell_0 = q$. Additionally, for every other trait $t \in T \setminus \{0\}$, $\ell_t = 0$. If the collection of feasible selections is defined as in (5), then determining whether there exists a feasible selection is NP-complete.*

PROOF OF LEMMA 6. We use a reduction from the Independent Set Problem. An instance of the Independent Set Problem is given by a graph $G = (V, E)$ and an integer k . An *independent set* of graph G is a set S of vertices such that no two vertices in S are adjacent. The goal is to determine whether there exists an independent set containing at least k elements. Given an instance of the Independent Set Problem, we construct an instance of the Selection Problem with upper quotas as follows: create one individual for each vertex, and one trait (with only two individuals) for each edge. Let each trait have an upper quota of 1, and let the overall lower quota be k . Then for any selection, the upper quotas are satisfied if and only if the set of vertices associated with the selected individuals is independent. Therefore, deciding whether there exists a feasible selection is equivalent to determining whether there is an independent set of size at least k . $\square$

LEMMA 7. *Suppose there is a trait $0 \in T$, such that $\mathcal{I}_0 = \mathcal{I}$ and $u_0 = q$. Additionally, for every other trait $t \in T \setminus \{0\}$, $u_t = |\mathcal{I}|$. If the collection of feasible selections is defined as in (5), then determining whether there exists a feasible selection is NP-complete.*

PROOF OF LEMMA 7. We use a reduction from the Set Cover Problem. An instance of the Set Cover Problem is given by a ground set $\mathcal{U}$, a collection $\mathcal{S}$ of subsets of $\mathcal{U}$, and an integer k . A *cover* for $\mathcal{U}$ is a subcollection $C \subseteq \mathcal{S}$ of sets whose union is $\mathcal{U}$. The objective is to determine whether there exists a set cover $C \subseteq \mathcal{S}$ for $\mathcal{U}$ containing at most k elements. Given an instance of the Set Cover Problem, we construct an instance of the Selection Problem with lower quotas as follows: create one trait for each element in $\mathcal{U}$, and one individual for each set in $\mathcal{S}$ (having traits identified by the set). There is a lower quota of 1 for each trait, and an overall upper quota equal to k . Then a selection of individuals satisfies all minimum quotas if and only if the corresponding subsets cover $\mathcal{U}$. Therefore, determining whether there exists a feasible selection is equivalent to determining whether there is a set cover with at most k elements. $\square$

C PROOFS FROM SECTION 5

PROOF OF THEOREM 2. We define an invariant that the algorithm maintains, and show that any selection satisfying the invariant respects priorities. By Proposition 1, any selection that respects

priorities must be an outcome-based selection rule. To show that the invariant is maintained by the algorithm at the end of every step, we use a backward induction argument on the gender parity requirement. For this purpose, it will be convenient to relax the gender parity requirement and allow the gap between the number of male and female electees to be at most K for some $K \geq 1$. (Naturally, Algorithm 2 should be modified to swap as long as the gender gap is larger than K ; the rule in the Chilean election sets $K = 1$.)

Let $\mathcal{F}_K$ be the collection of feasible sets for the Chilean election problem under the condition that the gender gap is required to be at most K . That is, every set $S \in \mathcal{F}_K$ satisfies the following two properties: (i) the number of candidates in S from party i is exactly E_i ; and (ii) the difference between the number of men in S and the number of women in S is at most K . Clearly, $\mathcal{F}_K \subseteq \mathcal{F}_{K+1}$, for all $K \geq 1$.

Algorithm 2 begins by selecting the top E_i candidates from each party i . If this initial selection satisfies the gender-disparity requirement, the algorithm halts. Otherwise, exactly one gender is over-represented; we call it the majority gender. Each subsequent swap reduces the gender gap by exactly two, so the majority gender is fixed throughout the algorithm's execution. Note that the swap step always replaces a majority-gender candidate with a minority-gender candidate from the same party; the algorithm 'skips' parties only when looking for a replaceable majority-gender candidate, but the swap itself is always within a party. We assume that $\mathcal{F}_1$ is nonempty so that there is always a feasible selection for any K . Thus, a valid swap exists at each step of Algorithm 2.

Our proof proceeds by showing that Algorithm 2 maintains the following invariant throughout its execution: for each unselected candidate j belonging to party i , either

- (a) E_i higher priority candidates from party i are selected; or
- (b) j is a majority-gender candidate and the following conditions are both true:
 - (i) Every selected lower priority candidate belonging to party i is a minority-gender candidate.
 - (ii) If a lower priority *majority-gender* candidate j' belonging to party $i' \neq i$ is selected, every minority-gender candidate from party i' with lower priority than j' is also selected.

Satisfying Invariant at Termination $\implies$ Respecting Priorities. Let S be the final selection, which satisfies the invariant. Let j be an unselected candidate from party i . To show that S respects priorities, we must show that there is no feasible set S' containing j and all candidates in S with higher priority than j . This is immediate if condition (a) of the invariant holds because at most E_i candidates from party i can be selected. Suppose condition (a) does not hold; then condition (b) of the invariant must be satisfied. In particular, j is a majority gender candidate, and at least one lower-priority candidate from party i is selected. Thus, j must have been replaced by a lower priority minority-gender candidate in the swap step of the algorithm to satisfy the gender parity constraint. Consider any selection that includes every selected candidate with higher priority than j as well as candidate j . To satisfy the party quota constraint, at least one lower-priority candidate from party i must be *unselected*, but every such candidate is a minority candidate, and doing so will increase the gender gap by 2. Thus, to satisfy the gender parity constraint, some majority-gender candidate with lower priority than j , and so necessarily belong to a party different from i , must be swapped for a minority-gender candidate of the same party. By condition b(ii) no such choice exists.

Inductive Proof of the Invariant. We next prove that Algorithm 2 satisfies the invariant after every step of its execution. This is established by induction on K : For K large enough, the initial selection of the algorithm (picking the top E_i candidates from each party i) will satisfy the gender disparity requirement, and so (a) is satisfied for all unselected candidates. This establishes the base case for the induction.

Suppose the claim is true for $K \geq k$. We want to show that the claim remains true for $K = k - 1$. Let S_k be the selection found by this algorithm when the gender gap is required to be at most k . If $S_k \in \mathcal{F}_{k-1}$, then the claim is clearly true (as $\mathcal{F}_{k-1} \subseteq \mathcal{F}_k$). Otherwise, $S_k \notin \mathcal{F}_{k-1}$; i.e., the gender gap in S_k is exactly k .

Suppose S_k includes more men than women. (The case in which S_k includes more women than men is symmetric.) This implies that at least one of the men selected in S_k must be unselected and at least one woman not selected in S_k must be selected. Suppose that the algorithm in the swap step replaces man m with woman w , both belonging to party i . We argue that the selection S_{k-1} obtained by dropping m and adding w maintains the invariant. This swap (in fact, every swap) drops a majority-gender candidate and adds a minority-gender candidate, so properties b(i) and b(ii) are unaffected for any candidate¹¹. By the induction hypothesis, the invariant is satisfied at S_k . This establishes the invariant at S_{k-1} for all unselected candidates from parties other than i , and also for all unselected candidates from party i with higher priority than m or lower priority than w .

We next prove the invariant for m , the candidate unselected in the current step of the algorithm. From the algorithm's choices, we know that m is the lowest priority selected man from party i in S_k and w is the highest-priority woman from party i not chosen in S_k ; this establishes b(i) for m at S_{k-1} . Now, suppose $m' \in S_k$ is a man with lower priority than m . From the algorithm's choice of m over m' , m' and m belong to different parties; moreover, m' was not a candidate during the swap step, which implies that every woman belonging to the same party as m' and having lower priority than m' was already selected; this establishes b(ii) for m at S_{k-1} .

We finally prove the invariant for the remaining unselected candidates from party i . Note that the priority for any such candidate j is below that of m and above that of w . By Algorithm 2's choice of w , j must be a man (otherwise j would have been chosen instead of w), and $j \notin S_k$. By the induction hypothesis j satisfied the invariant at S_k . If j satisfied b(i) and b(ii) at S_k , then j continues to do so at S_{k-1} as the swap does not affect these properties. Otherwise, j satisfied (a) at S_k , i.e., exactly E_i candidates higher priority from party i are chosen in S_k . In particular, S_k does not contain any lower priority candidate belonging to party i . As S_{k-1} is obtained from S_k by removing m and adding w , the only selected lower priority candidate from party i is w , whose is a woman. Thus, b(i) of the invariant is satisfied for j at S_{k-1} . Note that b(ii) of the invariant is also satisfied for j at S_{k-1} by the algorithm's choice of m as the candidate for swap.

□

¹¹These properties can be affected only if a minority-gender candidate is *unselected* or if a majority-gender candidate is *selected*.